



Annual Provider Education



HUNTSVILLE
HOSPITAL



MADISON
HOSPITAL

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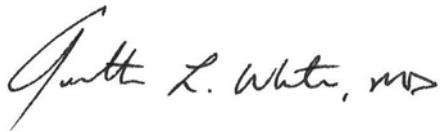
Dear Provider:

We need your help in preparation for our Joint Commission accreditation survey. Joint Commission standards work in step with our High Reliability “Journey to Zero” initiative to reduce patient harm. Joint Commission standards and the Centers for Medicare & Medicaid Services (CMS) conditions of participation require that providers are educated on a variety of topics. Experience has taught us that it is much better to be compliant during the survey than to be required to implement drastic measures post-survey.

Also of importance – CMS and the Centers for Disease Control and Prevention (CDC) require that hospitals report the rate of vaccination among the Medical Staff and all other healthcare team members. To help collect this information, please click on the link at the end of the e-book and answer the questions regarding your flu vaccination status this season. We appreciate your daily efforts in regards to patient care.

As always, if you have questions, concerns, and or ideas on ways to improve patient safety and care, please do not hesitate to give us a call.

Sincerely,



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Huntsville Hospital
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Our Mission

To provide high quality care and coordinated services that improve the health of our communities

Our Vision

To be the choice for care and careers in the communities we serve

Our Values

Safety, Compassion, Integrity, Excellence, Innovation, Accountability,
Diversity, Equity and Inclusion

2025-2026 MEDICAL STAFF LEADERSHIP



Dr. Paul B. Tabereaux
President



Dr. Jessica S. Branscome
Vice President



Dr. Jeffrey A. Walker
Secretary

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HOSPITAL PRIORITIES

Each year the hospital evaluates and revises priorities based on data and opportunities for improvement identified throughout the year. Each priority is aligned with organizational goals and is both process and/or outcome driven. Annually, the priorities are reviewed by multiple committees including Quality Council, Medical Executive Committee and ultimately approved by the Professional Affairs, Quality and Safety Committee, the quality committee of the Board. The priorities for this year are:

Organizational

Commit to Zero Harm

- Improve the patient identification process
- Reduction of Hospital-Associated Infections
- Reduction of preventable readmissions and preventable mortalities

Antimicrobial Stewardship

Pain Management (to include opioid safety and assessment/re-assessment)

Patient Safety

Capacity Management

Fall Safety

Environment of Care

Prevention of Workplace Violence

GOVERNING BOARD

The Healthcare Authority of the City of Huntsville is the governing board for Huntsville Hospital Health System.



Mike Goodman,
Chairman



Amit Arora, MD
Vice-Chairman



Frank Caprio,
Secretary/Treasurer



Philip W. Bentley, Jr.



Jim Bolte



Kerry Fehrenbach



Janice Johnson



Tharon Honeycutt



Michael McFadden, MD



Roy Rollings



Patricia Sims, EdD

PROVIDER EDUCATION

MEDICAL STAFF COMMITTEES

Committees of the Medical Staff are either standing or special. All committees and committee chairs are appointed by the President of the Medical Staff, with the exception of the MEC, Credentials Committee, and Nominating Committee, or as otherwise provided in the MS Bylaws. Unless otherwise specifically provided, a vacancy on any staff committee, with the exception of the Credentials Committee and Nominating Committee, will be filled in the same manner in which original appointment to such committee was made. A vacancy on the Credentials Committee and Nominating Committee will be filled by the MEC to serve out the remaining term.

Medical Executive Committee

Composed of the Medical Staff Officers, Department Chairs, and up to six practitioners representing Hospital service lines, and a practitioner who represents the Madison Hospital. The MEC Has the primary oversight authority related to professional activities and functions of the Medical Staff and performance improvement activities regarding the professional services provided by Medical Staff members and Allied Health Professionals with Clinical Privileges.

Credentials Committee

Composed of a nine members of the Active Medical Staff (six whom are elected by the General Medical Staff and three past-presidents). The Credentials Committee coordinates the credentialing and delineation of privileges process for Medical Staff members and allied health professionals.

Wellness Committee

Serves in an advisory capacity to the Credentials Committee, MEC, and Health Care Authority. The Committee oversees the monitors the practitioner's physical and/or mental health as it will affect staff membership and/or clinical privileges, as well as practitioner participation in the Alabama Physician Health Program.

Quality Council

Composed of members of the Medical Staff appointed by the President and representatives of hospital management. The Quality Council is responsible for oversight of the Quality Management Program, and initiates quality improvement work groups and teams as needed.

Medical Staff Affairs Committee

Composed of members appointed by the Medical Staff President. The Medical Staff Affairs Committee serves in an advisory capacity to administration, MEC and the Health Care Authority on matters related to Medical Staff Bylaws, policies, procedures, patient care policies, and patient education. The Committee also reviews recommendations from the Residency Advisory, Medical Records, Information Services and Continuing Medical Education Committees.

Medical Quality Management Committee

Composed of the chair (appointed by the Medical Staff President), vice chairs of the Medical Staff Departments, and other members at large as needed. The MQM oversees the Medical Staff peer review activity, review of conclusions and recommendations made by the Clinical Department Review Committees including recommending continuing education programs based on quality findings. This Committee also acts as a forum for discussion of quality of care issues identified in the Medical Staff Departments.

Clinical Department Review Committee (CDRC)

Composed of members appointed by the respective Department Chairs. The CDRC reviews the quality and appropriateness of patient care rendered in the inpatient and ambulatory setting by respective Department members.

Continuing Medical Education Committee

Composed of members appointed by the Medical Staff President. The CME Committee plans, implements, coordinates and promotes an ongoing CME program by providing CME opportunities based on the assessed needs of the Medical Staff.

Functions of the Medical Staff

The Medical Executive Committee acts as necessary to ensure that the functions of the Medical Staff are properly carried out. These functions include, but are not limited to: Blood Review, Tissue/Autopsy Review, Operative/Invasive Procedure Review, Medication Review, Analysis of Infection Surveillance, Medical Record Review, Resource Utilization, Safety and Environmental Risk Review, and the Cancer Program.

PHYSICIAN WELLNESS

The Medical Staff Wellness Committee provides for confidential consultation and support to physicians facing concerns related to alcoholism, substance abuse, behavioral or mental health issues, and/or physical illness. When necessary, the Committee works in collaboration with the Alabama Physician Health Program (APHP).

Concern regarding the wellness of a colleague should be reported / referred to the Medical Staff Wellness Committee (256-265-8858) or directly to the Alabama Physician Health Program (Phone 334-954-2596 or Toll Free 800-239-6272). The Alabama Code Sec. 34-24-361b (reporting law) requires that physicians report any other physician... “who may be unable to practice medicine with skill and safety to patients by reason of illness, drugs, inebriation... or as a result of any mental or physical condition.”

The Alabama Code Sec. 234-24-404 specifies that all information, interviews, reports, statements, memoranda or other documents are declared to be privileged and confidential. “APHP records shall not be available for court subpoena or for delivery proceedings.”

HUNTSVILLE HOSPITAL ANTIMICROBIAL STEWARDSHIP PROGRAM

The prompt initiation of antibiotics to treat infections has been proven to reduce morbidity and save lives. However, overuse and misuse of antibiotics has also contributed to the growing problem of antibiotic resistance, which has become one of the most serious threats to public health. Patients who are unnecessarily exposed to antibiotics are placed at risk for serious adverse events with no clinical benefit. Studies indicate that 30-50% of antibiotics prescribed in hospitals are unnecessary or inappropriate. Misuse occurs in healthcare settings for a variety of reasons including: use of antibiotics when not needed, continued treatment when no longer necessary, use of broad-spectrum agents to treat susceptible bacteria, and wrong antibiotic to treat an infection. Therefore, improving the use of antibiotics is an important patient safety and public health issue as well as a national priority.

Huntsville Hospital has developed an Antimicrobial Stewardship Program (ASP) to implement standardized, evidence-based antimicrobial stewardship processes to optimize antimicrobial therapy. The goals of the ASP are to promote appropriate use of antimicrobial agents, optimize clinical outcomes and patient safety, impede antimicrobial resistance, reduce adverse drug events, prevent secondary infections, and minimize healthcare costs without adversely impacting quality of care. The ASP at Huntsville Hospital is implemented through the coordinated efforts of the Antimicrobial Management Team, Anti-Infective Sub-Committee of the Pharmacy and Therapeutics Committee, and Infection Control.

The ASP works to accomplish these goals through varied means involving medical staff from diverse disciplines. In order to facilitate stewardship practices during the prescribing process, the ASP has developed use criteria for certain antimicrobials, updated and developed order sets, and published Huntsville Hospital's annual antibiogram. The ASP also continually evaluates new antimicrobial agents for formulary inclusion, conducts medication use evaluations, as well as manages antimicrobial backorders. It is through these measures that the ASP at Huntsville Hospital is able to reach the ultimate goal of optimizing antimicrobial therapy to improve patient outcomes, impede antimicrobial resistance, and maximize healthcare resources.

Top 6 Accomplishments of the Antimicrobial Stewardship Program: 2021-2022

- Implemented an extended-infusion policy for cefepime and meropenem to improve dose-optimization
- Created PowerPlans for the following infectious diseases: community-acquired pneumonia, urinary tract infections, skin and soft tissue infections, and *Clostridioides difficile* colitis
- Implemented a new stewardship process for Nasal MRSA PCR screening for patients with a diagnosis of pneumonia and receiving an anti-MRSA antibiotic
- Ensured preparedness for The Joint Commission's New Antimicrobial Stewardship Standard
- Updated PowerPlans with regard to surgical prophylactic antibiotics based on the results of the most recent antibiogram
- Implemented a microbiology comment nudge in the setting of asymptomatic bacteriuria

ANESTHESIA SERVICES AND SEDATION/ANALGESIA GUIDELINES

Standards and expectations for all patients receiving anesthesia services, including but not limited to, Topical or Local Anesthesia, Minimal sedation, Moderate sedation, Monitored Anesthesia Care (MAC), Regional Anesthesia and General Anesthesia at Huntsville Hospital can be found in the *Anesthesia Services and Sedation/Analgesia Guidelines*. The guidelines apply to all locations in the Hospital including all departments on all campuses and off-site locations where anesthesia services are provided.

Anesthesia involves administration of medication to produce a blunting or loss of pain perception (analgesia), voluntary and involuntary movements, autonomic function, memory, and/or consciousness. In contrast, analgesia is the use of medication to provide pain relief through blocking pain receptors in the peripheral and/or central nervous system where the patient does not lose consciousness but does not perceive pain.

Anesthesia exists along a continuum. For some medications there is no bright line that distinguishes when their pharmacological properties bring about the physiologic transition from the analgesic to the anesthetic effects. Furthermore, each individual patient may respond differently to different types of medications.

The following define the levels of sedation/analgesia and anesthesia based on *CMS 42 CFR §482.52 - Condition of participation: Anesthesia services* and the American Society of Anesthesiologists' definitions and practice guidelines:

- Topical or Local Anesthesia: the application or injection of a drug or combination of drugs to stop or prevent a painful sensation to a circumscribed area of the body where a painful procedure is to be performed. There are generally no systemic effects of these medications, which also are not anesthesia, despite the name.
- Minimal Sedation: a drug-induced state during which patients respond normally to verbal commands. Although cognitive function and physical coordination may be impaired, airway reflexes and ventilatory and cardiovascular functions are unaffected. This is also not anesthesia.
- Moderate Sedation/Analgesia ("Conscious Sedation"): a drug-induced depression of consciousness during which patients respond purposefully (reflex withdrawal from a painful stimulus is NOT considered a purposeful response) to verbal commands, either alone or accompanied by light tactile stimulation. No interventions are required to maintain a patent airway, and spontaneous ventilation is adequate. Cardiovascular function is usually maintained. Moderate or Conscious Sedation is not defined as anesthesia.
- Monitored Anesthesia Care (MAC): anesthesia care that includes the monitoring of the patient by practitioner who is qualified and privileged to administer anesthesia. Indications for MAC depend on the nature of the procedure, the patient's clinical condition, and/or the potential need to convert to a general or regional anesthetic. Deep sedation is included in MAC.

Deep Sedation is a drug-induced depression of consciousness during which patients cannot be easily aroused but respond purposefully (reflex withdrawal from a painful stimulus is NOT considered a purposeful response) following repeated or painful stimulation. The ability to independently maintain ventilator function may be impaired. Patients may require assistance in maintaining a patent airway, and spontaneous ventilation may be inadequate. Cardiovascular function is usually maintained

- Regional Anesthesia: the delivery of anesthetic medication at a specific level of the spinal cord and/or to peripheral nerves, including epidurals and spinals and other central neuraxial nerve blocks, is used when loss of consciousness is not desired but sufficient analgesia and loss of voluntary and involuntary movement is required.

- General Anesthesia: a drug-induced loss of consciousness during which patients are not arousable, even by painful stimulation. The ability to independently maintain ventilator function is often impaired. Patients often require assistance in maintaining a patent airway, and positive pressure ventilation may be required because of depressed spontaneous ventilation or drug-induced depression of neuromuscular function. Cardiovascular function may be impaired

Clinical privileges in anesthesiology are granted to physicians and other providers qualified to administer anesthesia whom are qualified by training to render patients insensible to pain and to minimize stress during surgical, obstetrical and certain medical procedures, as defined in the Medical Staff credentialing standards. Clinical privileges are also granted to practitioners who are not anesthesia professionals to administer sedative and analgesic drugs to establish a level of moderate or minimal sedation, as defined in the Medical Staff credentialing standards.

Anesthesia Services reviews adverse events, medication errors and monitors quality and safety indicators related to anesthesia, and to the administration of medications in clinical applications that the Hospital has determined to be sedation/analgesia rather than anesthesia. Cases will be referred to the Department of Anesthesia Clinical Department Review Committee (CDRC) and other CDRCs as appropriate.

Minimal Sedation

Any area in which minimal sedation is given must have appropriate equipment and an ACLS, PALS, and/or NRP certified RN readily available, in the event sedation moves to the moderate level. Minimal sedation and local anesthetics may be administered by a qualified and privileged Anesthesia Provider, a Non-Anesthesiologist Sedation Practitioner, a Sedation RN, Advanced Practice Nurse (APN), or Physician's Assistant (PA) in compliance with hospital policies and guidelines to administer prescribed sedating and analgesic medications and monitor patients during minimal sedation. A Sedation RN may only work under the direct supervision of a properly trained and privileged physician.

The supervising doctor is responsible for all aspects involved in the continuum of care – pre-, intra-, and post-sedation. While a patient is sedated, the supervising doctor must be immediately available, and is primarily responsible for pre-procedure patient evaluation. Sedation professionals may decline to participate in specific cases if they feel uncomfortable in terms of any perceived threat to quality of care or patient safety.

Moderate Sedation

Moderate sedation may be administered by a qualified and privileged Anesthesia Provider, a Non-Anesthesiologist Sedation Practitioner, a Sedation RN, APN, or PA. Sedation RNs, APNs, and PAs may only work under the direct supervision of an appropriately privileged physician. Physicians and dentists who hold clinical privileges to administer moderate sedation may supervise the administration of moderate sedation. Sedation RNs, APNs, and PAs may decline to participate in specific cases if they feel uncomfortable in terms of any perceived threat to quality of care or patient safety.

All providers of moderate sedation are required to have appropriate privileges and competencies. Moderate Sedation/analgesia for diagnostic, therapeutic, and invasive procedures is practiced throughout the hospital in accordance with the following guidelines.

- If the patient is on a general nursing unit or other area not appropriate for Moderate Sedation/Analgesia, the patient is transferred to an appropriate area or the appropriate staff and equipment is brought to the patient's bedside. In either case, the patient is monitored pre-, intra-, and post-sedation in accordance with these guidelines.
- The Anesthesia Provider or Non-Anesthesiologist Sedation Practitioner ordering the sedation/performing procedure is in the procedural area and ready to start the procedure prior to the initiation of sedation/analgesia.
- The person monitoring the Sedation/analgesia cannot be the same person who performs the procedure unless the procedure is itself that of monitoring (i.e. EEG).
- The individual managing and monitoring the patient receiving procedural sedation/analgesia has no other responsibilities during the procedure that would leave the patient unattended or compromise continuous monitoring.

Monitored Anesthesia Care (which includes Deep Sedation)

Due to the significant risk that patients may enter a state of general anesthesia, MAC, which includes deep sedation, will be administered only by appropriately privileged practitioners as follows:

1. NPO status: Patients must be NPO to receive MAC or deep sedation. NPO status will lead to canceling of the case or a significant delay of the case.
2. Procedural sedation can be performed initially by the physician with deep sedation privileges. Once stable sedation and adequate monitoring are established by the physician, the nurse can monitor the patient while the physician performs the procedure.
3. For patients who have had procedural sedation, a post sedation evaluation is performed once the patient is sufficiently recovered from the acute administration of the sedation so as to participate in the evaluation, e.g. answer questions appropriately, perform simple tasks, etc.

General and Regional Anesthesia

General and regional anesthesia may be administered only by anesthesiologists and supervised CRNAs.

PRE-SEDATION PROTOCOL:

Patient Selection:

- A. If the patient has an ASA score of 4 or greater, consider Monitored Anesthesia Care (MAC)
- B. Satisfactory arrangements for transportation of outpatients after the procedure are made before the patient is sedated. (The patient is released with a responsible person.)

Pre-sedation Assessment:

- A. A pre-sedation assessment and reassessment by the Anesthesia Provider or Non-Anesthesiologist Sedation Practitioner are required prior to administration of Moderate Sedation/Analgesia. Then both assessments are documented in the patient's medical record with the reassessment reviewed/completed with respect to the patient's condition immediately prior to administration of Moderate Sedation/Analgesia by the Anesthesiologist or Non-Anesthesiologist Sedation Practitioner supervising the sedation who signs the pre-sedation assessment form.
- B. The pre-sedation assessment may be combined with or may be separate from the pre-procedure History and Physical. The requirements for the pre-procedure History and Physical are outlined in the Contents of Medical Records policy.
- C. A pre-sedation assessment includes but is not limited to:
 1. Physical status assessment (review of systems, vital signs, airway, cardiopulmonary reserve).
 2. ASA status.
 3. Past and present drug history including drug allergies.
 4. Previous adverse experience with local medications, sedation and analgesia, regional and general anesthesia, or family history of anesthesia complications.
 5. Results of relevant diagnostic studies.
 6. History of tobacco, alcohol, and substance use/abuse.
 7. Verification of patient NPO status.
 8. Plan and choice of sedation.

D. Multiple Administrations of Moderate Sedation/ Analgesia

In patients who are undergoing procedures that require Moderate Sedation/Analgesia multiple times per day or single/multiple times on successive days, the initial pre-sedation assessment is sufficient as long as the physician performing the procedure determines through subsequent evaluation of the patient that no change has occurred in patient's clinical status that might alter the outcome of administration of sedation/analgesia. A short note is placed in the patient's medical record stating that an evaluation of the patient's clinical status has been performed prior to subsequent administration of Moderate Sedation/Analgesia.

Informed Consent

- A. Prior to the administration of sedation/analgesia, the patient or the authorized person is informed by the Anesthesia Provider or Non-Anesthesiologist Sedation Practitioner about the risks, benefits, and alternatives of Sedation/Analgesia and consents to the proposed sedation plan.
- B. The Anesthesiologist or Non-Anesthesiologist Sedation Practitioner documents this informed consent discussion in the medical record and the Permit for Sedation/Analgesia is signed by the patient or authorized person.

NPO Guidelines

- A. Verification of NPO status in all patients occurs before the start of sedation/analgesia and the following guidelines are followed unless the Anesthesia Provider or Non-Anesthesiologist Sedation Practitioner has weighed the benefits for a shorter NPO period and it is documented in the medical record.

Type of Food	Examples	Acceptable Time Frame
Clear Liquids	Liquids you can see through such as water, apple juice, coffee (no creamer), Pedialyte, other clear juices (no pulp), plain jello. Last drink may not be more than 1 regular cup (8 oz.)	2 hours before the scheduled surgery/ procedure time
Breast Milk		4 hours before the scheduled surgery/ procedure time
Infant Formula		6 hours (for infants under 1 year of age) before the scheduled surgery/ procedure time
Light Meal	Typically consists of toast and clear liquids. No fried or fatty foods and no meat.	6 hours before the scheduled surgery/ procedure time
All other foods & liquids	All solids, all milk, all meat containing products, all fried foods, all cheeses, ice cream, alcohol, and illicit drugs	8 hours before the scheduled surgery/ procedure time

Children at risk for regurgitation or aspiration (e.g., known gastro-esophageal reflux, extreme obesity) may benefit from pharmacological therapy to reduce gastric volume and increase gastric pH before sedation or from a longer NPO period of time prior to the procedure. If delayed gastric emptying is present, an Anesthesiology or GI consult is considered before sedation/analgesia.

Patients may take morning medications with a sip of water only. Diabetic medications should not be taken on the morning of surgery. Blood Pressure Medications considered an ANGIOTENSIN II RECEPTOR BLOCKER (ARB) or ANGIOTENSIN CONVERTING ENZYME (ACE) INHIBITOR should not be taken within 8 hours prior to the scheduled surgery/procedure time. Herbal medications need to be discontinued 10-14 days prior to surgery. Verify with surgeon the need to discontinue any blood thinners.

B. Radiological Procedures: Certain radiological procedures require the administration of oral fluids in conjunction with sedation/analgesia for both Adults and Pediatrics. Risk of aspiration during these procedures is weighed against the benefits of sedation/analgesia.

C. Emergency cases: It is noted that pain and narcotics often reduce gastric motility, which increases the risk of aspiration.

MALIGNANT HYPERTHERMIA REVIEW

Malignant Hyperthermia Syndrome is a rare, genetically transmitted, life-threatening complication that may be triggered by drugs commonly used in anesthesia and rapid sequence intubations. CMS requires that education and training be provided to any staff who may be involved in the care of a patient with malignant hyperthermia. Triggers for malignant hyperthermia are not only anesthesia gases, but also succinylcholine, which is used by CRNAs, Emergency Medicine Physicians, Intensivists, Hospitalists, Residents, and Pulmonologists for rapid sequence intubation.

The syndrome begins with a hypermetabolic state in muscle cells and is thought to involve altered mechanisms of calcium function at the cellular level. Characteristics of the syndrome include:

Tachycardia (sudden, unexplained)	Hyperkalemia
Tachypnea, spontaneous ventilation	Cyanotic mottling of skin
Profuse sweating	Arrhythmias
Myoglobinemia	Unstable blood pressure
Discolored urine	Elevated CPK
Fasciculations and/or rigidity	Hypercapnia
Dark blood in surgical field despite adequate inspired oxygen	Metabolic acidosis
Fever - rapid rise, sustained to as high as 108° F (42.2° C) or more	Respiratory acidosis

A Code MH is called for any suspected malignant hyperthermia crisis in non-procedural areas, where anesthesia is not already present. Code MH activation brings a response team and the MH cart to help manage the crisis. MH carts must be available within 10 minutes and are located in the following areas:

HH Main - Old PACU	Orthopedic & Spine Tower - 2nd floor PACU	W/C - OR Hallway
GMT - PACU	Madison - Anesthesia Workroom	

Treatment

In case of MH emergency Contact the 24-HOUR MH HOTLINE **800-644-9737**.

Remove/turn off triggering agents

Hyperventilate with 100% oxygen >10Lpm.

Administer Ryanodex (dantrolene) 2.5 mg/kg IV rapidly through a large bore IV; reconstitute each 250mg vial with 5mL sterile preservative free water for injection. Repeat until signs of MH are reversed; sometimes more than 10mg/kg (up to 30mg/kg) is necessary. **For patients <5kg.** Give Dantrium (dantrolene) 2.5mg/kg IV; reconstitute 20mg vial with 60ml sterile preservative free water for injection. Dantrium (dantrolene) and neonatal dosage chart are located in a labeled bag in Women's & Children's MH cart. Repeat until signs of MH are reversed.

Administer Bicarbonate 1-2 mEq/kg, if blood gases are not yet available.

Draw labs ASAP ABG, CBC, BMP, CK, Urine myoglobin, PT/INR, PTT

Cool the patient surface, NG lavage, open wound, rectum, IV. Stop cooling once temp decreases to 38°C (100.4°F).

Dysrhythmias usually respond to treatment of acidosis and hyperkalemia. Use standard drug therapy *except calcium channel blockers, which may cause hyperkalemia or cardiac arrest in the presence of Ryanodex(dantrolene).*

Hyperkalemia treat with hyperventilation, bicarbonate, glucose/insulin, calcium.

Bicarbonate 1-2 mEq/kg IV.

Pediatrics 0.1 units insulin/kg and 1ml/kg 50% glucose;
adults 10 units regular insulin IV and 50 ml 50% glucose.

Calcium chloride 10mg/kg or calcium gluconate 10-50 mg/kg for life-threatening hyperkalemia.

Check glucose levels hourly.

Place foley catheter, arterial line, multiple large bore IV's, consider Central Venous Line or PA monitoring as needed.

Follow ETCO₂, electrolytes, blood gases, CK, continuous core temperature monitoring, urine output and color, coagulation studies. If CK and/or K⁺ rise more than transiently or urine output falls to less than 0.5ml/kg/hr, induce diuresis to >1ml/kg/hr and give bicarbonate to alkalinize urine to prevent myoglobinuria-induced renal failure.

When stable, transfer the patient to the ICU for at least 24 hours. Key indicators of stability include: ETCO₂ is declining or normal, heart rate is stable without dysrhythmias, hyperthermia is resolving, muscular rigidity is resolving. Place “Post Acute Phase Malignant Hyperthermia Orders” on chart.

Documentation of MH crisis is completed on the Code MH flow sheet and placed under “Other” tab of chart. Add volatile anesthetics and succinylcholine to patient’s list of allergies. A safety event report should be completed on any MH crisis.

Succinylcholine Administration

Patients in non-procedural areas receiving succinylcholine for Rapid Sequence Intubation (RSI) should be monitored for 2 hours after the administration of succinylcholine for signs and symptoms of MH. This includes monitoring core temperature, muscle rigidity, and urine color. Remember that the intubation itself can cause tachycardia and hypertension. These signs alone do not indicate MH. Obtaining ETCO₂ value will help determine if a MH Crisis is occurring. Respiratory Therapy has a portable ETCO₂ monitor if the ICU does not have the capability to obtain this reading.

EMERGENCY MEDICAL TREATMENT AND LABOR ACT

EMTALA Fact Sheet: Physicians and Qualified Medical Personnel (QMP)

EMTALA is a federal statute, which preempts state laws or hospital policy that requires Medicare-participating Hospitals with Emergency Departments to screen and treat the emergency medical conditions of individuals in a non-discriminatory manner, regardless of their ability to pay, insurance status, national origin, race, creed or color. Its provisions apply to all patients, and not just to Medicare patients.

EMTALA also applies to a specialty-hospital without an Emergency Department if they have capacity to treat, (i.e. behavioral or psych unit)

- 1. Primary Rule** - if any individual, or person on their behalf, comes to the emergency department seeking emergency care, (as a layperson would define emergency) the hospital must provide an appropriate Medical Screening Exam (MSE) to determine if an Emergency Medical Condition (EMC) exists.
 - The phrase “Comes to the Emergency Department” is not confined to the Emergency Department alone, but includes any area on the hospital campus, including the hospital-owned ambulance service
- 2. Medical Screening Exam (MSE)**
 - The MSE should be completed by a physician or qualified medical personnel (QMP) as determined by hospital bylaws or other board-approved documents.
 - Triage is not a substitute for an MSE – the Triage assessment only determines the order in which patients receive the MSE.
 - The MSE must be documented in the medical record.
 - The MSE is necessary to determine if an Emergency Medical Condition (EMC) exists – and can only be determined by the physician’s or QMP’s assessment – not the hospital’s protocols
 - Once the MSE has been completed, it is permissible to complete the registration process and obtain payer information.
- 3. Emergency Medical Condition (EMC)**
 - If an EMC exists, the hospital must initiate stabilizing treatment within the hospital’s capability and capacity, or arrange an appropriate transfer to an accepting hospital in accordance with EMTALA provisions
- 4. EMC Disposition**
 - If the EMC is stabilized and/or resolved:
 - The individual is discharged home with follow-up instructions – EMTALA obligations end; or
 - The individual is admitted to the hospital for further stabilizing treatment – EMTALA obligations end.

- Definition of Stabilized: with respect to an EMC, that no material deterioration of the condition is likely, within reasonable medical probability, to result from or occur during the transfer of the individual from the facility or, in the case of a woman in labor, that the woman delivered the child and the placenta.
- If the EMC is Un-stabilized and the patient requires treatment outside of the hospital's capability – the hospital will arrange for an "appropriate transfer" to an accepting hospital.

5. Appropriate Transfer

- Before implementing a transfer of an individual with an un-stabilized EMC, the hospital is required to provide stabilizing treatment within its capability that minimizes the transfer risk to the individual and, in the case of a pregnant woman in labor, the health of the unborn child
- EMTALA does not apply to the transfer of Stable patients; however, if the patient is unstable, the hospital may not transfer the patient unless:
 - i. A physician certifies the medical benefits from the transfer outweigh the risks of the transfer; or
 - ii. An individual requests the transfer in writing after being informed of the hospital's obligations under EMTALA and the risks associated with transfer.

6. Four Requirements of an Appropriate Transfer:

1. The transferring hospital provides medical treatment within its capacity that minimizes the risks to the individual's health and, in the case of a pregnant woman in labor, the health of the unborn child.
2. The receiving facility has available space and qualified personnel, has agreed to accept the transfer of the individual, and to provide appropriate medical treatment.
3. The transferring hospital should send all necessary medical records related to the emergency condition to the receiving facility that is available at the time – test results received after transfer should be sent to the receiving hospital consistent with HIPAA provisions.
4. The transfer is effected through 'qualified personnel' with appropriate transportation equipment, including the use of medically appropriate life support measures. The physician at the sending hospital (not at the receiving hospital) has the responsibility to determine the appropriate mode, equipment and attendants for transfer.

7. Other Notable Facts About EMTALA:

- Hospitals with specialized capabilities are obligated to accept transfers from hospitals who lack the capability to treat unstable emergency medical conditions.
- EMTALA does not apply to scheduled outpatients, or patients presenting at off campus outpatient clinics that do not routinely provide EMC, or inpatients of the hospital.
- Patients who are in "Observation Status" are not considered inpatients, even if they stay overnight – therefore, EMTALA still applies and a receiving hospital should accept them if they have the capability required for treatment of the EMC.
- EMTALA Violation Penalties:
 - i. Civil Monetary Penalties (CMP) can be imposed upon a hospital that violates EMTALA: Hospitals with 100+ beds = \$50,000 per violation/< 100 beds = \$25,000
 - ii. Individual Physicians can be fined by CMS for violations of EMTALA, and both the hospital and physician can be excluded from the Medicare/Medicaid programs.

8. Documentation Expectations:

- Document to substantiate appropriateness of actions, (i.e. H & P, Test Results, etc.)
- Document appropriate actions such as the Transfer Certification Form or Refusal of Treatment forms as indicated;
- Ideally, each ED patient's record would indicate the disposition of the EMC (i.e. did not exist, stabilized, resolved, discharged, transferred or admitted)
- Documentation of patient disposition releases or ends the EMTALA obligations for the hospital and physician.

QUALITY AND SAFETY EDUCATION

CORE MEASURES

What is a Core Measure?

A core measure is intended to measure or evaluate the process of medical/surgical care. These nationally determined and publicly reported measures show how often patients receive a treatment or intervention that has been shown by scientific evidence to improve outcomes of specific medical conditions or surgical procedures. Health care experts and researchers are constantly evaluating the evidence to make sure that the measures and guidelines are kept current. Core Measures are one way to compare the quality of care provided at hospitals.

What Core Measures are we tracking?

Sepsis (SEP)

Venous Thromboembolism (VTE)

How can doctors impact Core Measures?

Knowledge of these essential “core” elements of care and coordination of care is an important part of providing excellent care for our patients. Physicians have the potential to identify opportunities to improve the process of care and bring them to our attention. By working together we promote excellence.

SEPSIS CORE MEASURE

Bundle Up quick!

The Sepsis Core Measure consists of a **three and six hour bundle**. In the **first three hours** of presentation of severe sepsis we must:

- Draw an Initial Lactate Level
- Collect blood cultures
- Administer Broad Spectrum Antibiotics

If the patient meets the criteria for Septic Shock*

** Patient is hypotensive or has a lactate level is ≥ 4*

Administer fluid resuscitation bolus with 30 mL/kg of crystalloid fluids. This bolus must be infused within 3 hours of presentation! May use Ideal Body Weight if patient has a BMI > 30 . May give a lesser volume bolus for heart failure, renal failure, or not fluid responsive with appropriate documentation of reason for lesser volume.

In the **first six hours** of presentation of severe sepsis/septic shock we must:

- Draw a repeat lactate level
- Administer Vasopressors for persistent hypotension (2 consecutive low BPs) in one hour following the end of the IV crystalloids.
- Complete repeat volume status and tissue perfusion assessment* if the patient met criteria for septic shock

**Volume status and tissue perfusion assessment must be performed and documented by LIP.*

Volume Status and Tissue Perfusion Assessment must consist of either:

A focused exam including **ALL** of the following:

- Vital Signs
- Cardiopulmonary Exam
- Capillary Refill Evaluation
- Peripheral Pulse Evaluation
- Skin Examination

OR any two of the following four:

- Central Venous Pressure Measurement
- Central Venous Oxygen Measurement
- Bedside Cardiovascular Ultrasound
- Passive Leg Raise or Fluid Challenge

QUALITY IMPROVEMENT

The quality improvement methodology used by the organization is Plan-Do-Check-Act (PDCA).

Plan Identify the opportunity for improvement; define your objective. Ask why are we doing this and how can we do it differently to make it better. Develop a multidisciplinary team; identify what will be measured.

Do Collect and analyze the data. Implement change strategies.

Check Review all measurement data. Determine the effects of the implementation on the process.

Act Analyze the differences to determine their causes. The analysis will allow investigators to determine where to apply changes that will render improvement. When the results of the PDCA do not render an improved state, the scope is redefined and the PDCA cycle is applied again until there is improvement.

Although we primarily use PDCA for improvement initiatives, when appropriate, the organization has resources that allow for the use of Lean Sigma, Six Sigma and other improvement methodologies.

Quality priorities & initiatives

Huntsville Hospital has chosen to also adopt the Join the Journey and Chasing Zero campaigns to address efforts in reducing Hospital Acquired Infections and other potentially preventable events as our overarching priorities. In addition to our long-term priorities, the organization has chosen several patient safety topics.

- | | |
|--|---------------------------------|
| – Elimination of Patient Identification Errors | – Capacity Management |
| – Reduction of Preventable Readmission | – Anti-microbial Stewardship |
| – Prevention of Hospital Acquired Infections | – Workplace Violence Prevention |
| – Fall Safety | |

Each priority is aligned with organizational goals and is both process and outcome driven. Priorities are reviewed by multiple committees including Quality Council, Medical Executive Committee and ultimately the Professional Affairs, Quality and Safety Committee of the Healthcare Authority of the City of Huntsville.

SAFETY ORIENTATION

Huntsville Hospital Safety Officer

Chad Ridings, Hospital Safety Officer

Office Location: Quality Department (2nd floor Main hospital)

Email: chad.ridings@hhsys.org

Office number: (256) 265-6904 | Cellular number: (256) 656-8018

Reporting an emergency from a hospital phone: call 4-5555

Reporting an emergency from an off-site phone: call 9-911

Emergency Overhead Pages/Announcements

- **CODE BLUE:** Cardiac or Respiratory Arrest
- **CODE RED:** Fire
- **CODE RED ALL CLEAR:** Fire all clear
- **CODE PINK:** Infant abduction
- **CODE AMBER:** Any missing person
- **CODE GRAY:** Need for security personnel
- **CODE BLACK:** Bomb threat
- **CODE GREEN ALERT:** Disaster plan stand-by alter.
- **CODE GREEN ACTIVATE:** Disaster plan activation
- **OVERHEAD ANNOUNCEMENT IN PLAIN LANGUAGE**
 - Active shooter or armed assailant
 - Severe weather
 - Any emergency that requires special response (i.e. evacuation)

Safety Reminders

- **RACE:** Rescue, Alert, Confine, Extinguish or Evacuate
- **PASS:** Pull pin, Aim, Squeeze handle, Sweep base
- **FIRE EXTINGUISHER & PULL STATION LOCATION:** Near stairwell or exit
- **STANDARD PRECAUTIONS:** Treating everyone as potentially infectious
- **BLOOD & BODY FLUID PRECAUTIONS:** Always wear gloves when handling blood or bodily fluids
- **SPIL:** Secure, Protect, Inform, Leave
- **MED-ALERT:** Medical emergency (non-patient)
- **MED-ALERT TRAUMA:** Medical emergency involving uncontrollable bleeding of a visitor or staff
- **SDS (Safety Data Sheets):** For information on any material or chemical spill call (256) 265-MSDS (6737)
Information you need when calling:
 - Product name / Manufacturer name
 - Universal Product Code (UPC), if available
 - Your department fax number

Electrical Safety Guidelines

Before use of equipment, inspect electrical cords/equipment and immediately notify plant operations at **5-2700** for damaged or defective cords and/or equipment.

Back Safety

Lift equipment is available for patient handling. Use the safe lifting position when picking up boxes, equipment or heavy objects. Be sure to squat by bending at the hips and knees, keeping the body in a straight line. Your feet should be shoulder width apart and slightly turned out to center your weight. Bring object lifting close to your body with elbows and arms tucked in close to your body. Let your legs do the lifting. Do not twist your body.

RAPID RESPONSE

During a hospitalization, patients may experience a worsening condition requiring an evaluation by a critical care team. This evaluation is called a rapid response. Huntsville Hospital's Rapid Response Teams are made up of critical-care nurses and respiratory therapists highly trained in emergency situations to provide urgent care for patients experiencing a serious change in their medical condition including chest pain, choking, difficulty breathing, shortness of breath or possible seizure activity.

Caregivers may activate the Rapid Response Team by calling 4-5555 or by pressing the patient's Nurse Call button. Patients or family members who notice a serious change may press the Nurse Call button to immediately voice their concern to a Patient Response Center representative. The request is then immediately dispatched to all nursing staff assigned to the patient's unit. The patient is quickly assessed to determine whether or not to activate the Rapid Response Team.

Scope

The Rapid Response Teams are assigned to support all patients. All other areas off-site locations receive support by calling HEMSI (9-911 from a Huntsville Hospital phone).

The goals of the rapid response program are to:

- Decrease the number of cardiopulmonary arrests
- Decrease patient mortality
- Support floor nurses in critical care decision-making
- Improve patient outcomes
- Prevent deaths of patients outside intensive care settings

Initiation of a rapid response occurs when (but not limited to):

Airway

- Respiratory distress/shortness of breath
- Threatened airway
- Light headed/dizziness

Breathing (Adult only)

- RR > 30 per minute
- RR < 12 per minute

Circulation

- Symptomatic change in BP
- Chest Pain
- Altered Rhythm
- Symptomatic change in HR (sustained HR > 125 (Adult only) or HR < 45)
- Uncontrolled bleeding

Neurology

- New onset of altered mental status
- Change in level of consciousness

Other

- Intuitive concern about the patient
- Rapid deterioration in patient condition

The team is comprised of a critical care nurse and respiratory therapist.

They must:

- Consider the situation critical and respond promptly, in person
- Assess, if respiratory therapist is not needed, they are free to go
- Stabilize
- Assist with communication to physician, if needed
- Educate and support in a non-threatening approach
- Assist with transfer, if necessary
- Complete the RRT record

DO NOT initiate a rapid response for the following issues:

- Need an EKG interpreted
- Need a new IV
- Need help restraining or moving a patient
- When you have not fully assessed the patient: all vital signs, oxygen in place and turned on, IV therapy intact and infusing appropriately
- Inability to manage workload

HIPAA AND CORPORATE COMPLIANCE

HH Health System has adopted a set of values: integrity, excellence, innovation, accountability, compassion, safety, diversity, equity and inclusion. Integrity and honesty directs our dealings with people outside the hospital, including patients, families, businesses and government agencies. HH Health System is required to comply with all federal and state laws, especially those focused on preventing fraud and abuse and to report any behavior that may be considered illegal or unethical. Corporate Compliance is the method to ensure that each of us does what is right and trustworthy every time we deal with anyone. Employees not only have the responsibility for being honest, we have the responsibility for reporting concerns about possible violations of rules and policies. Employees who make a good faith report of wrongdoing will not suffer negative actions as a result of the report. Reports will be kept confidential as much as possible. Employees may also report safety and quality concerns to Joint Commission (JC). Some examples of fraudulent or wrongful acts include:

- Violating laws protecting patient privacy (HIPAA)
- Stealing money, property or anything of value
- Receiving pay or documenting time not worked
- Profiting from insider knowledge of the hospital, using your position to influence business or soliciting personal gifts from hospital contractors
- Any dishonest act
- Violation of a state professional licensure or other regulatory agency
- Misuse of electronic, social or other media that violates hospital policy and/or patient privacy
- Activities that could make up a false claim

Purpose

The Office of the Inspector General (OIG) lists the five most important Federal fraud and abuse laws that apply to physicians as the False Claims Act (FCA), the Anti-Kickback Statute (AKS), the Physician Self-Referral Law (Stark law), the Exclusion Authorities, and the Civil Monetary Penalties Law (CMPL). Violating these laws could result in criminal penalties, civil fines, exclusion from the Federal health care programs, or loss of your medical license from your State medical board.

The False Claims Act (FCA) exists to fight fraud, or false claims, against the federal government. A false claim may take many forms, including, for example, overcharging for a product or service, failing to perform a service, delivering less than the promised amount or type of goods or services, underpaying money owed to the government, and charging for one thing while providing another. Other actions that could be considered a False Claims are: billing a patient for a medical procedure, service, or item that was not provided; billing for research that was not performed or, inappropriately changing or destroying any hospital record, medical or other. Applicability: In general, the FCA covers fraud involving any federal agency or program with the exception of tax fraud.

Damages and Penalties

Under the FCA, anyone who knowingly submits or causes the submission of false claims to the government is liable for damages of up to three times the erroneous payment plus civil penalties of up to \$25,076 per false claim. Criminal penalties for submitting false claims include imprisonment and criminal fines.

Mechanism

A private individual, called the whistleblower or relator, who reports information regarding false claims is authorized to file a case in federal court (on behalf of the government) those entities that committed fraud. These are called qui tam suits. The Department of Justice then decides (on behalf of the government) whether to join the whistleblower or relator in prosecuting these cases.

Whistleblower or Relator's Share

If the case is successful, the whistleblower may share in the recovery. The amount of the relator's share depends on multiple factors, including whether the relator planned and initiated the false claim.

Whistleblower or Relator Rights and Protections

The FCA provides a remedy for whistleblowers who are discharged, demoted, suspended, or “in any other manner discriminated against in the terms and conditions of employment by his or her employer” in retaliation for filing an FCA case. To receive the benefits of the employment protections of the FCA, courts generally require that an employee demonstrate that: (1) he or she was engaged in an activity protected by the statute in furtherance of a qui tam suit, (2) the employer knew of the employee’s qui tam actions, and (3) the employer retaliated against the employee because of those actions. If the court finds a whistleblower was terminated or otherwise mistreated for filing a qui tam lawsuit, the employee is entitled to reinstatement at the same level, two times the back pay owed plus interest, and compensation for any special damages, such as attorney’s fees, sustained as a result of the discrimination.

AKS is a criminal law that prohibits the knowing and willful payment of “remuneration” to induce or reward patient referrals or the generation of business involving any item or service payable by the Federal health care programs. Remuneration includes anything of value and can take many forms besides cash, such as free rent, expensive hotel stays and meals, and excessive compensation for medical directorships or consultancies.

Damages and Penalties: Criminal penalties and administrative sanctions for violating AKS include fines, jail terms, and exclusion from participation in Federal health care programs. Under the CMPL, physicians who pay or accept kickbacks also face penalties of up to \$50,000 per kickback plus three times the amount of remuneration. Safe harbors protect certain payment and business practices that could otherwise implicate the AKS from criminal and civil prosecution. To be protected by a safe harbor, an arrangement must fit squarely in the safe harbor and satisfy all of its requirements.

The Physician Self-Referral Law, commonly referred to as the Stark law, prohibits physicians from referring patients to receive “designated health services” payable by Medicare or Medicaid from entities with which the physician or an immediate family member has a financial relationship, unless an exception applies. Financial relationships include both ownership/investment interests and compensation agreements. The Stark law is a strict liability statute, which means proof of specific intent to violate the law is not required. The Stark law prohibits the submission, or causing the submission, of claims in violation of the law’s restrictions on referrals.

Damages and Penalties: Penalties for physicians who violate the Stark law include fines as well as exclusion from participation in Federal health care programs.

Exclusion Statute: the OIG is legally required to exclude from participation in all Federal health care programs any individual or entity convicted of the following types of criminal offenses: (1) Medicare or Medicaid fraud, as well as any other offenses related to the delivery of items or services under Medicare or Medicaid; (2) patient abuse or neglect; (3) felony convictions for other health-care-related fraud, theft or other financial misconduct; and (4) felony convictions for unlawful manufacture, distribution, prescription, or dispensing of controlled substances. The OIG has discretion to exclude individuals or entities on several other grounds.

Damages and Penalties: If you are excluded by the OIG from participation in the Federal health care programs, then Medicare, Medicaid, and other Federal health care programs, such as TRICARE and the Veterans Health Administration, will not pay for items or services that you furnish, order, or prescribe. Excluded physicians may not bill directly for treating Medicare and Medicaid patients, nor may their services be billed indirectly through an employer or a group practice.

CMPL: the OIG may seek civil monetary penalties and sometimes exclusion for a wide variety of conduct and is authorized to seek different amounts of penalties and assessments based on the type of violation at issue. Some examples of CMPL violation include: violating the FCA or AKS; violating Medicare assignment provisions; violating Medicare physician agreement; violating EMTALA regulations; or making false statements or misrepresentations on applications or contracts to participate in the Federal health care programs.

Damages and Penalties: Penalties range from \$10,000 to \$50,000 per violation.

Compliance with all legal and regulatory guidelines is the minimum standard of performance at HH Health System. We should strive to meet or exceed requirements. Employees are required to complete Corporate Compliance training and must sign an annual Affirmation Statement. HH Health System has a Department of Corporate Compliance and a Corporate Compliance Officer who supervises the Compliance Program. If you think someone has committed fraud or an illegal act, you are required to report it immediately to a manager, the Corporate Compliance Officer, or the Corporate Compliance Department. To make a report, notify compliance@hhsys.org or call (256) 265-9951. If you want to make an anonymous report you should call 1(800) 442-0959 24 hours a day, 7 days a week.

HIPAA: HEALTH INSURANCE PORTABILITY AND ACCOUNTABILITY ACT

The Health Insurance Portability and Accountability Act (HIPAA) is a Federal Law designed to protect patient information. It does so through the use of Privacy Standards and Security standards (electronic and physical). The Hospital has implemented a program covering HIPAA requirements. Training for HIPAA is required annually for all employees and is located in Net Learning. The program emphasizes the importance of privacy and security. Each employee, physician, volunteer and student has a responsibility to safeguard Protected Health Information (PHI) in any format: spoken, written or electronic. PHI includes all information about patients and can be as basic as the patient's name or other personal information. Protected Health Information should be accessed or shared only when necessary to do your job. Approved use of PHI is limited to treatment of the patient, payment for treatment and operations of the hospital. PHI should be disposed of in approved receptacles in non-public areas.

If you overhear someone discussing PHI, you should keep this information to yourself. HH Health System has assigned a Privacy Officer to ensure that privacy is maintained. If you believe that a violation of privacy or unauthorized disclosure of PHI has occurred, you should report that to your manager or call the Privacy Officer at (256) 265-9257 or privacyofficer@hhsys.org. The Privacy Officer will investigate every report of a privacy violation.

Ensuring patient privacy is the responsibility of every employee. You can do your part by adhering to the following basic requirements:

- Only access patient information when you need it to treat the patient or providing services for payment. Other uses of Protected Health Information (such as hospital operations) will be covered in your job description.

- You shall not access the patient records of your family members, co-workers or anyone other than provided for in your job responsibilities even with permission from the patient. This is a violation of Huntsville Hospital policy. Records may be obtained from the Medical Records department by accessing the hospital website with authorization from the patient.

- HIPAA allows patients to receive a list of all disclosures of their PHI. Therefore, the hospital monitors and audits access to patient records.

- Always use care when discussing PHI with another caregiver. Make sure the other person has a job-related reason for the information. Whenever possible, make arrangements so that your conversation is not overheard.

Security of Protected Health Information:

- Obtain (written or verbal) approval from your manager or director before taking PHI to an offsite location. Department leaders should develop an alternative process to ensure security of PHI off-site.

- Do NOT store PHI on any portable electronic device unless it is encrypted or otherwise protected by the Information Technology (IT) Department's Security Officer.

- Do NOT store PHI on personal electronic devices unless authorized by the IT Security Officer.

- Do NOT store PHI on any local computer hard-drive ("C drive") unless approved by the IT Department's Security Officer.

- Notify the Security Officer and Compliance Officer immediately of any loss, or suspected loss, of an electronic device that may have PHI stored on it (desktop computer, flash drive, camera, phone, laptop/notebook or any portable device).

- Violations of the Health Insurance Portability and Accountability act (HIPAA) policy may result in disciplinary action, up to and to include termination.

CMS CAHPS PROGRAM

What is the CAHPS Program?

The Centers for Medicare & Medicaid Services (CMS), through the Consumer Assessment of Healthcare Providers and Services (CAHPS) program- develop, implement and administer several different patient experience surveys. These surveys ask patients (or in some cases their families) about their experiences with, and ratings of, their health care providers and plans, including hospitals, home health care agencies, hospice, doctors, and same day surgical treatments, among others. The surveys focus on matters that patients themselves say are important to them and for which patients are the best and/or only sources of information. CMS publicly reports the results of its patient experience surveys, and some surveys affect payments to CMS providers.

CAHPS surveys are an integral part of CMS' efforts to improve healthcare in the U.S. Some CAHPS surveys are used in Value-Based Purchasing (Pay for Performance) initiatives. These initiatives represent a change in the way CMS pays for services. Instead of only paying for the number of services provided, CMS also pays for providing high quality services. The quality of services is measured clinically, administratively, and through the use of patient experience of care surveys.

Experience is not the same as satisfaction

Patient experience surveys sometimes are mistaken for customer satisfaction surveys. They do not focus on amenities. Patient experience surveys focus on how patients experienced or perceived key aspects of their care, not how satisfied they were with their care. Patient experience surveys focus on asking patients whether or how often they experienced critical aspects of health care, including:

- communication with their doctors
- understanding their medication instructions
- the coordination of their healthcare needs

Why is the Patient Experience Important?

- Better patient experiences are associated with better patient safety, improved clinical outcomes and higher patient satisfaction scores. This could be something as direct as the relationship between cleanliness and infection prevention. But it could equally be more tangential, such as the relationship between a patient feeling less anxious and being more receptive to aspects of their health care. Improved health outcomes can be associated with shorter stays and reduced readmissions.
- Many aspects of improved patient experience are also associated with greater equity in access to care. For example, providing translation services or services focused on supporting specific needs of older patients.
- Patient experience and staff engagement are intertwined, making it important to understand the connection between patients and staff. Being able to provide a positive experience is associated with improved staff morale and motivation, which can help support both recruitment and retention of staff.
- Positive patient experiences tend to lead to patients feeling more empowered to take a fuller role in their own healthcare. That could manifest in many ways, from being better able to understand their healthcare pathway, to being more prepared to speak out, enabling clinical staff to respond to any worries and concerns.
- Many of the factors that improve patient experience tend to be directly and indirectly associated with the same factors that can reduce risk. For example, something as apparently simple as an effective cleaning round will reduce slip and trip hazards.

What can I do to improve the Patient Experience?

- **Strengthen RN/Provider relations** — Nurses spend much more one on one time with the patients, so utilize their skills. Neither of you can function efficiently or effectively without the other. Foster a collaborative and supportive team.

- Convey Courtesy and Respect by utilizing the **Universal Relationship Tools**:
 - **Smile and Greet Others - Use the 10/5 Rule:** Meet anyone you encounter (patients, visitors, co-workers) with warmth and a kind greeting.
 - Smile and acknowledge others at 10 feet. Verbalize a greeting at 5 feet.
 - Be aware of body language - regardless of whether you are inside or outside of the room. How we act in both public spaces and private patient rooms creates the culture of our facility.
 - **Ask for and use preferred names.** The absence of introductions can instill confusion and anxiety and can diminish trust. Tell them your name and role, even if you think they know it or it's implied. Example: "Hello Mrs. Jackson, my name is John Smith and I will be your pulmonologist — which means lungs are my specialty. You can call me Dr. Smith. What do you like to be called?"
 - **Listen with Empathy and Intent to Understand:** Make eye contact, nod and ask clarifying questions when appropriate. Try to really understand why they might be feeling that way or asking that question. If we understand where patients are coming from we can better address their concerns.
 - **Communicate the positive intention of your actions:** This takes only a fraction of a second to say but communicates a much more caring intent. Examples:
 - Instead of "Here is your blanket," try "We got you a warm blanket, we want you to be comfortable."
 - Instead of "**scrubbing sanitizer in silence**" say "Let me wash my hands, so we can keep you safe from the spread of germs."
 - **Provide opportunities for others to ask questions:** Patients are often intimidated and feel helpless in medical situations. It is important that they know we have time to answer their questions and address their concerns.
 - Ask open ended questions like "What questions do you have?" This implies it is normal to have questions and we are happy to answer them!
 - **Sit whenever possible:** Changing planes can make it seem like a transition to a new conversation. Stand with introductions, sit during history, stand for exam and sit in a chair to discuss treatment plan. The patient will perceive four different parts of the exam and encourage the perception that "I have time for you."
 - **Explanation of care**
 - Provide diagnosis in a clear order
 - Share the diagnosis or what you are attempting to clarify or rule out and how you plan to do that. (Patients want to know what they have. Let them know the name and write it out for them. Equally important is explaining the serious diseases they DO NOT have, patients often assume the worst.)
 - Share the recommendations for treatment
 - Share the expected clinical course of the condition
 - Explain what the patient needs to do
 - Address symptoms management.

For more information on how you can help us improve our HCAHPS scores contact the Patient Experience team at extension 5-9449.

PATIENT ADVOCACY

What is a Patient Advocate?

A communication specialist that acts as a liaison between patients and their healthcare team providing information and conflict resolution to aid patients/family members in receiving better medical care, leading to better patient outcomes.

When should staff call a Patient Advocate?

If you need a liaison to help you resolve the patient or family members concerns or if the patient or family member requests to see or speak with a patient advocate.

How do you contact Patient Advocacy?

Patient Advocacy can be reached at (256) 265-9449.

Office of Patient Experience and Advocacy hours:

Monday – Friday 8 a.m. - 4:30 p.m.

If you need an advocate at night, weekends or holidays, contact the House Supervisor and they will act in the capacity of the advocate.

Complaints & Grievances - What's the difference?

Complaints...

- A concern or issue that can be resolved “*on the spot*”
- Verbal communication regarding concerns that would have routinely been handled if staff had been informed during the stay.
- Billing or privacy issues that do not include patient quality of care issues.
- All lost-and-found/parking issues
- If the concern could have been resolved by staff present if known about, this would still be considered a complaint not a grievance.

Grievances...

- A patient ask for a complaint to be filed as a “*Formal Complaint*”
- Any complaint that is communicated in writing or includes a request for a written response, or a letter attached to a satisfaction survey or comment card.
- Verbal or written complaints that involve patient abuse, neglect, patient harm, or hospital compliance with CMS Conditions of Participation
- Any complaints regarding a *quality of care issue*, disagreement with a coverage decision, billing issues, or they wish to appeal a premature discharge.
- Any concern filed directly with a regulatory agency i.e. State, JC, CMS

Allegations of abuse by a patient against staff – Correct steps to ensure patient safety and proper documentation are achieved

Policy Procedure

If an allegation of abuse or neglect has been made by a patient against an employee, appropriate care shall include:

- Separating the caregiver from the alleged victim. If the employee has direct patient contact, he/she will be immediately removed from direct patient care by the supervisor or anyone in the supervisor's chain of command. Actions are taken to ensure that a patient who has reported allegations of abuse against an employee still receives necessary care.
- Immediate medical, nursing, and psycho-social assessments are performed. The providers who perform the assessments enter findings in the medical record.

- An allegation of abuse against an employee by a patient is considered a grievance and should follow the Patient Complaint and Grievance Policy by immediately contacting the Unit Director, Service Line Administrator, Service Line VP, and Patient Experience & Advocacy.

Notification Procedure

- Incidents of actual or suspected neglect or abuse is to be reported and investigated using this process as a general guideline.
- Take immediate action to protect, comfort and assure the treatment and safety of the patient.
- Notify a supervisor or charge nurse
- A supervisor immediately notifies the Unit Director, Administrative (House Supervisor) or Administrator on call of the allegation of patient abuse. In the absence of the Unit Director, the employee immediately notifies the Service Line Administrator and VP.
- A Safety Event Report (SER) is entered into the HRP reporting system. The Service Line Vice President or department head immediately notifies Quality Management of the allegation of patient abuse.
- The Patient Advocate or designee is notified as soon as possible by the Unit Director, Administrative (House Supervisor), Administrator on call, Department Head or Service Line VP and this incident will follow the Patient/Visitor Complaint and Grievance policy. All feedback is to be entered into HRP and Quality Management will forward into the rL6 tracking system as appropriate.
- The direct supervisor and/or Service Line Vice President activate the chain of command appropriate to the area where the allegation has been reported.
- Allegations of patient abuse are reported to hospital Security who will alert the appropriate local authorities, as necessary

HIGH RELIABILITY ORGANIZATIONS

Not a Sprint but a Journey

For years, certain industries have been regarded as being highly reliable due to their success and ability to design processes that prevent catastrophic events. These organizations have been able to sustain a high safety record for extended periods of time. In the book, *Managing the Unexpected* by Weick and Sutcliffe, high reliability organizations are identified as those that “operate under very trying conditions all the time and yet manage to have fewer than their fair share of accidents.” The management of nuclear power plants has been noted for years to be high reliability organizations.

So, what about health care? Why does the health care industry continue to have a growing number of medical errors? Research conducted by Health care Performance Improvement, LLC differentiated between system and individual failures. System failures lead to individual failures which result in inappropriate acts. Seventy-seven percent of errors are due to failures in critical thinking, compliance and/or competency. The other 23% revolves around fatigue, inattention, distraction and ineffective communication.

What sets other industries apart from the health care industry? The simple answer is that safety is not just a priority. For high reliability organizations, it's a core value that is understood and embraced by everyone. They do not try to hide failures but celebrate them as gifts... an opportunity to prevent a catastrophic event. They are successful because they focus on five key principles to construct, flourish and maintain safety cultures. Those principles are:

- Preoccupation with failure – Actively seeking to know what we don't know
Reluctance to simplify interpretations – Continually digging until the root cause is known
- Sensitivity to operations – Constantly focusing on processes which lead to new operational initiatives
- Commitment to resilience – Always prepared on how to respond to events and insistent on rapid problem solving that prevents catastrophes
- Deference to expertise – The person with the most knowledge of the task is most valued

So how do we become a high reliability organization? We must be willing to embrace failure as a gift. We cannot be satisfied with the “simple” solution understanding the complex nature of health care. We must value the voice of the one that completes the task. We must be willing to do things differently. Being a High Reliability Organization is not a designation but a culture. Join us as we continue on our Journey to Zero - “Be Safe. Be Caring. Be Kind.” — to fulfill our vision “to be the choice for care and careers in the communities we serve.”

JOURNEY TO ZERO

UNIVERSAL RELATIONSHIP TOOLS

Universal Relationship Tools are simple communication practices to show respect and create familiarity among team members. Universal Relationship Tools increase frequency of communications. Some Universal Relationship Tools are non-verbal such as smiling and making eye contact. Some Universal Relationship Tools are verbal – meaning using words. Saying hello is a verbal Universal Relationship Tool.

Universal Relationship Tools are good factors in maintaining a culture of teamwork. Frequent use of Universal Relationship Tools creates an environment that – when we need to – we can think well together as a team.

5 Universal Relationship Tools

Respect at ALL times:

1. Smile and Greet Others - Use the 10/5 Rule
2. Ask for and use preferred names
3. Listen with empathy and intent to understand
4. Communicate positive intention of your actions
5. Provide opportunities for others to ask questions

UNIVERSAL RELIABILITY TOOLS

Universal Safety Tools are practices which improve human performance, both as teams and as individuals. Much more than communication tools, these strategies help us critically think, remember critical steps in hand-offs, escalate problems and request help. Together, these tools improve the reliability of human performance.

SAFETY BEHAVIORS <i>"I am accountable for and commit to being S.A.F.E."</i>	RELIABILITY TOOLS
Supporting each other	• Cross check to reduce errors • Cross coach using 5:1 feedback • Speak up using ARCC (Ask a question, Request a change, voice a Concern, and Chain of command)
keeping my Attention on task	• Self-check with STAR (Stop, Think, Act and Review)
Focusing on best practices	• Questioning attitude with Question and Confirm • Know Why and Comply with policy, procedure and checklist
Effectively communicating	• 3-way repeat back/read back • Phonetic and numeric clarification • Clarifying question • Handoff with the ISHAPED checklist • SBAR (Situation, Background, Assessment and Request)

RELATIONSHIP TOOL: SUPPORTING EACH OTHER

Cross Check to reduce errors

Cross Checking is the habit of keeping your attention on the people, equipment and environment around us, allowing instant recognition of problems.

Use Cross Checking at all times; keep an eye and ear out for trouble. If you suspect there a problem, speak up.

If you are uncomfortable speaking plainly to someone else use Speak-Up using ARCC as a way to point out an unsafe situation.

Cross Checking has several positive effects:

- Shared situational awareness, giving the advantage of more eyes and ears to catch things you might not notice on your own.
- You can ask for a check or a second opinion, giving the advantage of detecting unintended slips or lapses, critical thinking problems and even knowledge or skill deficiencies.
- Team members can reinforce the best practices in each other, and also discourage unsafe practices in each other in a positive way.

Cross Coach using 5:1 Feedback

Cross Coach Using 5:1 Feedback is a method for increasing the amount and quality of feedback that you give to others. There are two types of feedback – positive reinforcement and negative reinforcement:

- Positive, or encouraging, reinforcement makes a person more likely to repeat a behavior.
- Negative, or discouraging, reinforcement makes a person less likely to repeat a behavior.

When you see good practice, give your peer a quick nod, smile, “thanks” or, if you feel comfortable, a short compliment on their good practice. Non-verbal reinforcements are good because they are quick, easy and always taken as sincere. If you see a poor practice habit, look a little put-off or shake your head, and if you feel comfortable, ask “may I point out something to you?” then explain what is the correct practice.

The technique is called Cross Coaching Using 5:1 Feedback because you are most helpful to your peers when you average five positive reinforcements for every one negative reinforcement. Most people are closer to 1:1. This means most people need to increase the number of positive reinforcements. Active Cross Coaching in this ratio builds trust in relationships over time.

Speak up using ARCC

A sk a question
R equest a change
voice a C oncern
C hain of command

Use the Speak up for Safety ARCC Tool when you are uncomfortable speaking plainly about a safety concern. Start with a simple question. If the question doesn’t draw their attention to the problem, request a change. Quickly explain why and hand the dialogue back to them by adding “what do you think?” If the request did not change their thinking, use the Safety Phrase, “I am concerned that...” If nothing changes still, you have Chain of Command, which you can use to check your thinking and help you advocate for safety.

RELIABILITY TOOLS

Keeping my Attention on Task: Self-Check with STAR

S top: Pause for ONE second
T hink: About your act
A ct: Perform the act
R eview: Check for response

Use self-checking when processing information such as reading a label, flow-sheet, warning sign, posting, computer screen, indicator, etc. Use self-checking when doing things with your hands such as entering data, documenting, connecting devices, selecting supplies, selecting meds, etc. The one second pause keeps you thinking ahead of what you’re doing so that you never act without thinking.

Focusing on Best Practices: Questioning Attitude with Question and Confirm

Use Questioning Attitude every time you interpret information and every time you choose a rule from memory.

First, qualify the source of the information: is this source a good source for this information? Does this source have a history of being correct?

Next, question the information: does this information make sense? Is the information consistent with what you would expect?

Last, resolve anything that doesn’t make sense using an independent, qualified source when: the information is very important (high-risk), the information fails the source qualification or initial questioning tests, or the information appears to have changed from past experience.

Focusing on Best Practices: Know Why and Comply

Use Know Why and Comply when making choices based on policy, procedure and protocols. Review the rule and the reasons the rule exists. Apply Questioning Attitude. If the rule makes sense, then apply the rule in a way that meets both the letter and intent of the rule. Most policies, procedures and protocols are “written in blood” - errors were previously made that led to harm because we did not follow (or have) the rules.

Having a protocol in hand makes the user four times more reliable in applying that protocol. This is called continuous use. The protocol is physically with the user, the user reads and understands a step, performs the step and initials or signs for task tracking. Reference use is performing familiar tasks per policy, procedure or protocol from memory.

Effectively Communicate: Three Way Repeat/Read Back

1. Sender initiates using receiver's name.
2. Receiver acknowledges with "I understand..." and repeats (some reads) back verbatim.
3. Sender acknowledges with "that's correct."

Use repeat-back for oral communications where precise detail and meaning must be communicated. The proper response for sender acknowledgment is "that's correct." Do not say "right" because this can be interpreted as direction instead of an acknowledgment.

When performing a read-back, the receiver of information may ask the sender to "hold on for a moment, I need to write that down" while the receiver locates paper or signs on to the computer.

Phonetic and Numeric Clarification

A Alpha	H Hotel	O Oscar	U Uniform	Numeric Clarification: For 15 say "15, that's one – five" For 50 say "50, that's five – zero" For 0.9 say "0.9 that's zero-point-nine" For 3-4 say "the range of three to four"
B Bravo	I India	P Papa	V Victor	
C Charlie	J Juliet	Q Quebec	W Whiskey	
D Delta	K Kilo	R Romeo	X X-ray	
E Echo	L Lima	S Sierra	Y Yankee	
F Foxtrot	M Mike	T Tango	Z Zulu	
G Golf	N November			

The three best times to use phonetic clarification are when you are saying a patient name, drug name, or a procedure name. When using phonetic clarification say the name, then spell the name by saying the letter and the phonetic clarification for that letter.

Numeric clarification is even easier and saves more lives. The best times to use numeric clarifications are room number, dose, rate, and result. When using numeric clarification – say the number and then say the digits.

Effectively Communicate: Clarifying Questions

Always ask clarifying questions:

1. When in high-risk situations.
2. When information is incomplete.
3. When information is ambiguous.

Good thinking starts with Questioning Attitude. Ask the question in a polite, professional and helpful way. If asking a question of someone with high power distance, consider using ARCC to escalate safety concerns.

Effectively Communicate: Handoff with the ISHAPED Checklist

I ntroduction	Handoff Habits • You own it until you hand it off to an appropriate person. • If you accept a handoff for someone else, you own it until you hand it off to someone else. • Use repeat backs and ask clarifying questions.	Use a structured handoff tool like ISHAPED when providing a handoff communication to a team member about a patient, project or ongoing task transfer. Safe handoffs occur prior to a change in care provider or project owner and involves interactive, direct communications between care providers/project owners following a standard process. Ownership may be for an entire shift or part of a shift. Distractions must be minimized, and a checklist is used during the handoff process to ensure complete and relevant information, identification of potential risks and the opportunity to ask questions. The best clinical handoffs include the patient in the process.
S tory		
H istory		
A ssessment		
P lan		
E rror Prevention Review		
D ialogue		

EVENT REPORTING

What is a Safety Event Report (SER)?

A Safety Event Report is submitted any time there is a variance in care that adversely affected a patient or has the potential to affect the patient. The reports are submitted in the High Reliability Platform (HRP) located via PULSE. Safety Event Reports are categorized as a near miss, precursor or a serious safety event.

When should a SER be completed? - WHEN IN DOUBT, FILL IT OUT

Whenever there is an unexpected outcome

If the patient's course does not go exactly as planned

Whenever there is an unexpected outcome

SAFETY EVENT CLASSIFICATION (SEC)

Three steps of event classification

The HPI Safety Event Classification (SEC) is an outcome based event classification system involving three steps:

1. Determine if there were defects in care, called deviations from "Generally Accepted Performance Standards" (GAPS).

Deviations from GAPS are determined by comparing actual performance to expected performance. Internal practice expectations do not always reflect industry practice expectations in protecting workers from harm. Consideration of performance exceptions should include external as well as internal sources such as:

- Internal policies, procedures or protocols
- Nationally recognized best practices and standards of care
- Industry-imposed practice mandates and requirements
- Professional practice standards
- Objective review by other experts
- Organization's obligation to best protect the worker from harm

If defects and deviations from GAPS exist:

2. Establish if there was a cause-and-effect relationship between the deviations and the outcome.

While causation often is readily apparent, a cause-and-effect relationship between deviations and outcome may be difficult, and sometimes impossible, to determine. This may be the case when a patient is in critical condition or has other complications or co-morbidities (e.g., "the patient would have died anyway"). In such a case, the organization may be reluctant to declare a safety event or tend to downgrade the safety event classification. The ultimate goal of safety event classification and cause analysis is the identification and correction of root causes to prevent future events of harm.

The organization can safeguard against safety event under-classification by considering its obligation to do everything possible to provide an uncompromised, safe experience. Did the organization do its best to protect the patient from harm, regardless of the ability to definitely prove a direct case-and-effect relationship?

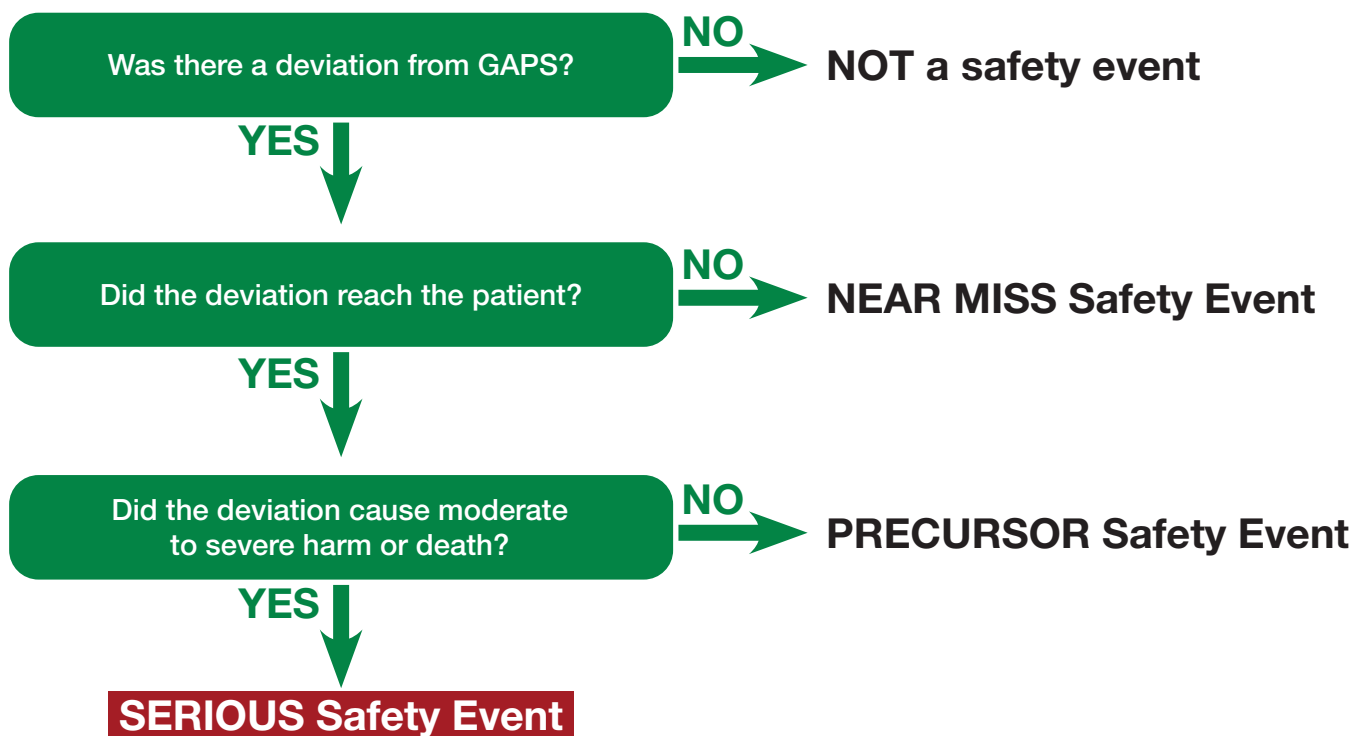
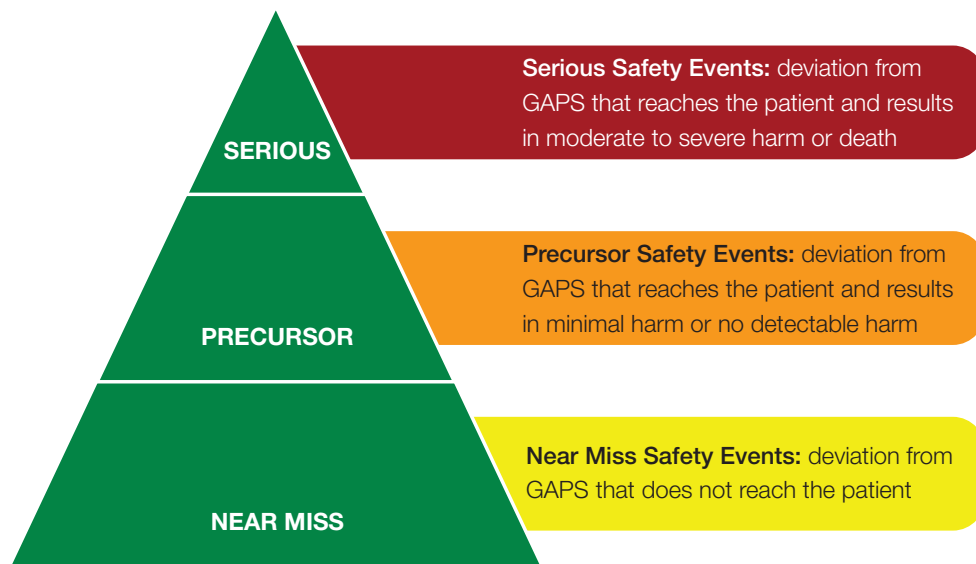
3. Classify the event according to the level of patient harm resulting from the event.

Assess the level of harm experienced by the patient and determine the safety event classification.

Serious Safety Event: Results in harm that ranges from moderate to severe patient harm or death

Precursor Safety Event: Results in minimal harm, no detectable harm or no harm

Near Miss Safety Event: Initiating error is caught before it reaches the patient by either a detection barrier built into the process, or sometimes by chance.



Best practice tips in safety event classification

- Identify a consistent group of people to serve as a “Safety Event Review Panel” to provide expertise, consistency and integrity in event classification. The group should be a mix of clinicians and methodology experts, and senior enough to gain organizational trust.
- When classifying an event, use the SEC algorithm and always ask all the questions - e.g. Was there a deviation? Did the deviation reach the patient? What was the level of harm?
- Charge one person with the responsibility for thinking/asking about precedent.
- Keep a record of challenging event classification causes and classification rationale. This record provides a useful reference when assessing similar future cases and enables the group to look at changes in their own perspective in event classification.
- What happens in the discussions, stays in the discussions. The group speaks with one voice outside the meetings.

The Known Complications test

A “known complication” is an adverse outcome supported in literature as a potential risk related to a procedure, treatment or test that is not present before the patient care encounter and occurs as a result of patient care.

If an event is perceived to be a known complication, the Known Complications test can be used to confirm the event as a complication and to determine if the providers did everything possible to prevent the negative outcome. When assessing a known complication, consider the following:

1. Was the procedure, treatment or test appropriate and warranted based on nationally recognized standards of care?
2. Was the complication a known risk, was it anticipated before the procedure, and was the standard of care applied to mitigate the risk?
3. Was the complication identified in a timely manner (i.e. at the time of the occurrence)?
4. Was the complication treated according to the standard of care and in a timely manner?

If the answer to all four questions is yes, the event is considered a known complication and not a safety event. If the answer to any question is no, the event is a safety event. Proceed in defining the classification based on the level of harm to the patient.

Safety event classification levels of harm

	Code	Level of Harm	Patient Harm Description (resulting from a deviation in GAPS)	Workplace Safety Harm Description
SERIOUS Safety Event	SSE 1	Death	Death	Event resulted in death
	SSE 2	Severe Permanent Harm	Critical, life-changing harm with no expected change in clinical status; includes events resulting in permanent loss of organ, limb or vital physiologic or neurologic function	Event results in 100% permanent, total disability in which the individual will no longer be able to perform work duties
	SSE 3	Moderate Permanent Harm	Significant harm with no expected change in clinical condition yet not sufficiently severe to impact activities of daily living or business functioning; includes events that result in permanent reduction in physiologic reserve, disfigurement and impaired or aided sense or function	Event results in a significant or permanent negative effect or there is evidence of physical injury requiring medical intervention and results in significant lost time or permanent restricted/transfer duty.
	SSE 4	Severe Temporary Harm	Critical, potentially life-threatening harm yet lasting for a limited time with no permanent residual; requires prolonged transfer to a higher level of care/monitoring, transfer to a higher level of care for a life-threatening condition, or an additional major surgery, procedure or treatment to resolve the condition	Event produced a significant temporary effect or there is evidence of physical injury requiring medical intervention and results in a greater than two-week period of missed or restricted/transfer duty.
	SSE 5	Moderate Temporary Harm	Significant harm lasting for a limited time; requires a higher level of care/monitoring or an additional minor procedure or treatment to resolve the condition	Event produced a significant temporary effect or there is evidence of physical injury requiring medical intervention and results in less than two-week period of missed or restricted/transfer duty.
PRECURSOR Safety Event	PSE 1	Minimal Permanent Harm	Minor harm with no expected change in clinical status; requires little or no intervention	Event produces minimal permanent harm. First aid or medical assessment is conducted but no restricted/transfer of duty.
	PSE 2	Minimal Temporary Harm	Minor harm lasting for a limited time only; requires little or no intervention	Event produces minimal temporary harm. First aid or medical assessment is conducted but no restricted/transfer of duty.
	PSE 3	No Detectable Harm	Reached the patient without ability to determine the existence or fact of harm, yet harm may exist; includes events where the onset of harm may occur later in time	Event does not produce visible evidence of physical injury and no first aid is required. The potential for future harm is possible but very unlikely.
	PSE 4	No Harm	Reached the patient but results in no harm, with sufficient information available to determine that no harm occurred	Event required no first aid, did not produce a negative effect, and there is no visible evidence of physical injury.
NEAR MISS Safety Event	NME 1	Unplanned Barrier Catch	Passed through all error detection barriers and did not reach the patient because it was caught by chance or by barrier not designed into the system	Caught or avoided by chance where no design process or barrier exists. No error detection barrier previously established.
	NME 2	Last Strong Barrier Catch	Passed through early error detection barriers and was caught by a last strong error detection barrier designed into the system	Caught or avoided when a last error detection barrier succeeded in stopping harm but a design process or early error detection barrier failed.
	NME 3	Early Barrier Catch	Caught by an early error detection barrier designed into the system's defense in depth	Caught by an early error detection barrier

CLINICAL EDUCATION

EMR DOWNTIME

When we are notified that our EMR system (1Chart) will be unavailable for end users, you will be made aware via an “e-blast” to all providers from the Medical Staff Office. The email will provide the following information:

- Date of the outage
- Timeframe
- Location of your respective patient lists.

Additionally, signs will be posted in all provider areas (lounges, dining room, parking garage). Feel free to always contact your IT Physician Network Team by calling the IT Help Desk at (256) 265-7777.

ANTICOAGULANTS

Anticoagulants are recognized as high risk/high alert drugs by TJC. TJC requires that anticoagulants be initiated and maintained via a protocol. Huntsville Hospital has implemented the following safe processes for these drugs:

Unfractionated heparin

Standardized protocols built as PowerPlans in 1Chart (ADULTS): when ordering, select the initial regimen (CARD, HOSP, or NEURO) and the adjustment sliding scale (Standard or Low, +/- boluses with adjustments)

Heparin guidelines: PTT monitored at least daily and platelets monitored at least every 3 days

Low molecular weight heparin

LMWH guidelines: anti-Xa levels and platelets monitored as appropriate

Renal adjustments: pharmacists adjust doses for renal dysfunction per Pharmacy and Therapeutics Committee approval

Warfarin

Warfarin guidelines: baseline INR required before initiation; INR monitored at least every 72 hours

Pharmacy management available by consult: Enter order to consult pharmacy for warfarin management; include target INR range and indication

Venous Thromboembolism (VTE)

VTE is a leading cause of preventable morbidity and mortality for hospitalized patients:

Approximately 1 million Americans develop VTE in the hospital or in the 30 days post-hospitalization annually

Please evaluate patients for VTE risk and order VTE prophylaxis, as appropriate

VTE Advisor in 1Chart prompts for VTE risk assessment and VTE Prophylaxis

Insulin Safety

Hypoglycemia due to insulin is the most common adverse drug event reported in the hospital. Common causes include failure to discontinue or adjust the dose when insulin requirements change, ‘dose- stacking’, and overly aggressive dose adjustments. Follow the tips below to reduce the risk of insulin induced hypoglycemia.

Consider adjusting scheduled insulin doses in the following situations:

Patient NPO, vomiting, low oral intake, or being transitioned from TPN/enteral feedings

Patient is being tapered off of corticosteroids

Patient has an episode of hypoglycemia

Avoid ‘dose-stacking’:

When contacted by a nurse regarding hyperglycemia, ask about the timing and doses of insulin administered in the last 4 hours

Consider the action of previously administered doses (especially of short and rapid acting insulin):

- Subcutaneous insulin lispro: peaks in 1-3 hours; duration of 3-5 hours
- Subcutaneous insulin regular: peaks in 1-5 hours; duration of 6-10 hours

Check the MAR before adjusting insulin orders:

Hyperglycemia may be due to ‘held’ insulin doses, rather than an insufficient ordered dose

Hypoglycemia may be due to ‘stacked’ rapid acting sliding scale doses, rather than an excessive dose of intermediate or long-acting insulin. This is especially common when a patient’s intermediate/long-acting insulin is not restarted in the hospital or when doses of intermediate/long-acting insulin are ‘held’ for low to low-normal glucose.

FALL PREVENTION & MANAGEMENT

“A patient fall is a sudden, unintentional descent, with or without injury to the patient, that results in the patient coming to rest on the floor, on or against some other surface (e.g., a counter), on another person, or on an object (e.g., a trash can). National Database of Nursing Quality Indicators (NDNQI®) counts only falls that occur on a eligible inpatient or ambulatory unit that reports falls.* When a patient rolls off a low bed onto a mat or is found on a surface where you would not expect to find a patient, this is considered a fall. If a patient who is attempting to stand or sit falls back onto a bed, chair, or commode, this is only counted as a fall if the patient is injured”(National Database of Nursing Quality Indicators).

**The inpatient or ambulatory unit is defined as an area where nursing care is provided, regardless of who is providing care for the patient at the time of the fall. It is best to determine the “footprint” of each unit to determine if the fall counts for the unit.*

NOTE: Lowering a patient to the floor/ground counts as a fall and is reported through the Quantros Safety Event Reporting System for tracking and trending purposes. Details of the patient fall and associated assessments (vital signs, neuro checks, skin assessment, etc.) are documented in the patient’s medical record. In 1Chart, the fall is documented in “Nursing Update.”

Nurse have several interventions if the patient is at risk for falls but physicians can do too.

Nurses are required to follow the following Post Fall Management Strategies:

Assess patient for injury and place on high risk fall precautions.

Notify patient’s practitioner immediately. Unless emergent, the patient is to remain on the unit until the patient’s practitioner is notified.

Notify patient’s family, if appropriate, as soon as possible and document.

Document the fall in the patient’s medical record. Documentation includes the time and place of the fall, contributing factors, Physician notification, injuries sustained and treatments for injuries. Also, the nurse completes a Safety Event Report.

RESTRAINT MANAGEMENT

Non-Behavioral Restraints

The use of a non-behavioral restraint is initiated to prevent displacement of medical devices or when impaired cognitive function interferes with medical treatment.

Huntsville Hospital does not allow PRN restraint orders

Only use restraints after least restrictive methods have been exhausted.

Physician order is required immediately following the initiation of non-behavioral restraints

Physician must examine the patient within 24 hours

Non-behavioral restraint orders are limited to a calendar day and a new order is required daily. Restraint order stickers are used for continuing restraints. The stickers are placed on the order sheet in the chart, the order sheet is flagged, and the physician must sign, date and time the order. Dictation number is also required.

Behavioral Restraints

Behavioral restraints are initiated to manage violent and/or self-destructive behavior that jeopardizes the immediate physical safety of the patient, a staff member or others.

A provider must sign orders and conduct an in-person face to face assessment within 1 hour of initiation of behavioral restraints.

Orders are time-limited based on age:

>18	4 hours
9-17	2 hours
<9	1 hour

Only health care professionals that have received behavioral restraint training and deemed competent may apply behavioral restraints. Appropriate training consists of education and competency.

Only use restraints after least restrictive methods have been exhausted. Patient should be released from restraints as soon as possible, when safe.

PRN OR STANDING ORDERS ARE NOT ACCEPTABLE FOR ANY TYPE OF RESTRAINTS.

PAIN MANAGEMENT AND OPIOID SAFETY

Strategies for safe and effective acute pain management

- Over 107,000 people died in the US from a drug overdose in 2021
- 1 in 8 opioid-naïve patients exposed to opioids for a surgical procedure become long-term users
- 70-80% of prescribed opioids are not used after surgery
- Longer duration of opioid therapy is associated with opioid abuse

Assess patient for risk of opioid associated adverse effects

- Evaluate patient for opioid tolerance
 - Tolerance = around the clock use for last seven days (oxycodone 30 mg/day or more)
- Identify risk factors for opioid induced respiratory depression
 - Sleep apnea (or snoring as an indicator of potential undiagnosed sleep apnea)
 - COPD
 - Renal dysfunction
 - Surgery late in the day (immediate recovery period during night-time hours)
 - Frail patients
- Assess for opioid abuse risk (Opioid Risk Tool or other)

Recommend non-pharmacological pain management options

- Before surgery, educate patients to expect some pain and that pain will be managed, but not eliminated
- Recommend distraction and relaxation techniques
- Heat and cold

Use non-opioid multimodal analgesia

- Multimodal regimens include doses of non-opioids before surgery
- Scheduled non-opioids after surgery reduce the need for opioids (continue for 3-5 days)
- Consider regional analgesia
- Consider acetaminophen, NSAIDs, celecoxib, gabapentin, tramadol and others
- Studies indicate the risk of bleeding with NSAIDs (including ketorolac) is very low

Use the lowest dose of opioids for the shortest period of time

- De-escalate quickly – opioid requirements decrease significantly by post-op day 1
- Avoid duplicate orders; order only 1 oral opioid (can be 1 tablet for moderate pain and 2 tablets for severe pain) and only 1 injectable opioid
- Discontinue injectable opioids as soon as patient able to take oral medications
- Start with tramadol 50 mg or Norco 5, unless patient takes higher doses at home

Prescribe a minimum number of opioid tablets at discharge

- Leftover opioids have helped fuel the opioid epidemic
- American College of Surgeons recommends basing prescription quantity on use the day before discharge
 - 0 opioid pills: do not prescribe opioids at discharge
 - 1 to 3 opioid pills: prescription quantity of 15
 - 4 or more opioid pills: prescription quantity of 30
- See OPEN (Opioid Prescribing Engagement Network) for recommendations by procedure

INFECTION CONTROL

Hand Hygiene

Hand hygiene continues to be the primary way to prevent the spread of infection. Hand hygiene should occur in between patient contact as well as upon entering and exiting the patient's rooms.

Alcohol based hand rubs should be used on a routine basis for hand hygiene: when hands are not visibly dirty such as before and after touching the patient and prior to placement of gloves for an invasive procedure and after removing gloves.

Hand washing using soap and water is warranted when caring for a patient with *Clostridium difficile* or when hands are visibly soiled or dirty.

Standard Precautions

Follow hospital Standard Precautions Policy: All human blood and body fluids are treated as if they were infectious for hepatitis B, HIV, and other blood borne pathogens

Use appropriate Personal Protective Equipment when it is anticipated that you will come into contact with patient's blood and body fluids.

Transmission Based Isolation Precautions

There are four types of transmission isolation precautions: Airborne, Contact, Droplet, and Contact Enteric. Whenever a patient is placed on isolation precautions, the isolation sign will be placed on the patient's door.

Please follow the appropriate precautions to prevent the spread of infection. The "safe zone area" within the door way can be utilized for patients on Contact and Contact Enteric Precautions. While within this area, please do not touch anything.

Patients must also be isolated upon suspicion or when diagnostic rule-out tests are ordered. Examples include: Contact Enteric precautions when testing for C. Difficile, Droplet precautions for Influenza, or meningitis, and Airborne precautions when ordering AFB cultures from a respiratory source.

Ordering AFB Cultures from a Respiratory Source

When a patient is suspected or confirmed of having M. tuberculosis, the patient must be placed on Airborne Isolation Precautions. This includes anytime an order is written for the testing of Acid Fast Bacilli (AFBs) from a respiratory source. Some examples of a respiratory source include the following specimens: sputum, lung tissue, or pleural fluid. When TB is not suspected, **it must be documented that TB is not suspected along with the diagnosis** that explains current condition. **Pneumonia is not an acceptable diagnosis** as TB is a type of pneumonia.

***Clostridium difficile* testing**

Patients should have 3 or more liquid stools in a 24 hour period before ordering C. Diff test. Formed stool will not be routinely tested. NOTE: Test of cure is not recommended. Lab will not retest a negative stool within 2 weeks nor retest a positive culture within 4 weeks. Isolation will be removed 2 weeks post resolution of symptoms.

Discontinuation of Isolation

Isolation Precautions may only be discontinued by an Infection Preventionist, to ensure compliance with hospital guidelines and policies to prevent the spread of infection.

Health Care Associated Infection (HAI)

Health Care Associated Infection means a localized or systemic condition that results from adverse reaction to the presence of an infectious agent or agents or its toxins and that was not present or incubating at the time of admission to the health care facility. A notification letter will be sent to Attending Physicians of those patients with reportable CLABSIs, CAUTIs, or SSIs. Please be advised, the criteria is based on surveillance definitions from CDC's National Health care Safety Network (NHSN) and not clinical definitions. Mandatory health care associated infection data shall be reported to Alabama Department of Public Health from the following categories:

Central line associated bloodstream Infections (CLABSIs) from the following critical care units within a health care facility:

- Adult Critical Care Units
- Pediatric Critical Care Units
- Neonatal Critical Care Units

Surgical Site Infections (SSIs) from the following procedures:

- Colon
- Hysterectomy- abdominal

Catheter-Associated Urinary Tract Infections (CAUTI) from the following patient care locations within a health care facility:

- General Medical Wards
- General Surgical Wards
- General Medical/Surgical Wards
- Adult Critical Care Units
- Pediatric Critical Care Units

Central Line-Associated Bloodstream Infections (CLABSI)

Our goal is to decrease the use of central lines.

Each year approximately 350,000 patients will acquire a catheter-related blood stream infection.

To prevent central line-associated bloodstream infections:

- Whenever possible, and not contraindicated, the subclavian site is preferred over the jugular and femoral site.

- Hand hygiene before and after contact with the patient or the central line.
- Maximal barriers for catheter insertion (mask, cap, sterile gown, and sterile gloves)
- Cover the patient with a full body drape during insertion
- Prep the patient's skin with chlorhexidine before catheter insertion
- Daily review of line necessity with prompt removal of unnecessary lines

Blood Culture Ordering and Collection Recommendations

General Principles regarding Bacteremia/Fungemia

- Blood cultures are indicated only when bacteremia or fungemia is suspected clinically.
- If bacteremia or fungemia is suspected, 2 sets of blood cultures should be drawn from separate venipuncture sites in adults (1 to 2 sets for children).
- Collection from any “plastic source device” (e.g. central venous catheter, PICC, port-a-cath, dialysis line) is strongly discouraged unless catheter-related infection suspected.
- Blood cultures should not be drawn from a peripheral IV.
- As standard blood cultures will readily detect fungemia due to *Candida* species, fungal blood cultures should only be ordered when non-*Candida* infections are suspected.

Catheter-Related Infections (e.g. including Central Venous Catheters, PICC lines, implanted port, hemodialysis catheter)

- a. In the event that a catheter-related bloodstream infection is suspected, evidence-based guidelines indicate that one set of blood cultures should be obtained from the catheter and one set via venipuncture.
- b. Collection via a catheter must be specifically ordered by a provider. If a peripheral venipuncture is not possible, two separate cultures obtained from the catheter should be collected 15 minutes apart.
- c. To document the clearance of a catheter-related infection, providers should wait to order blood cultures until the day after removal of the infected catheter (due to the risk of detection of transient bacteremia related to catheter removal).

Standing/Surveillance Orders for Blood Cultures

- a. Collection of blood cultures for routine surveillance in the absence of any clinical signs or symptoms (e.g. as a part of routine preoperative testing) is strongly discouraged.
- b. Routine standing or “as needed” orders for blood cultures are strongly discouraged. (e.g. daily cultures to document clearance of bacteremia or blood cultures x 2 for temp > 101 °F)
- c. In the absence of a suspected endovascular infection, collection of more than 2 sets of blood cultures in a 24 hour period is strongly discouraged.

Exceptions

If venipuncture collection is unsuccessful after multiple attempts, collection through a plastic source device will be allowed only with modification of the original blood culture order from a physician.

Surgical Site Infections (SSI's)

Surgical Site Infections

Surgical site infections occur in 1 to 3 out of every 100 surgical procedures.

To prevent surgical site infections, the following are best-practices:

- Administering pre-op antibiotics within 1 hour of the incision (2 hours for Vancomycin/quinolones)
- Administering the appropriate antibiotic for the procedure to be performed
- Discontinuing the antibiotic within 24 hours after surgery (48 hours for cardiac surgeries)
- If hair removal is required, hair is to be clipped. Razors are not used.
- Appropriate attire in the OR

- Appropriate surgical scrub prior to surgery
- Covering hair, including facial hair

Catheter Associated Bloodstream Infections (CAUTI)

Urinary Tract Infections- Foley Catheters

In order to reduce the population at risk for urinary tract infections, we need to limit the use of urinary catheters. Nursing assesses the patient daily for need to have the Foley using criteria below.

Criteria for urinary catheterization is as follows but not limited to:

- The Foley was placed/inserted by a Physician. MUST HAVE A PHYSICIAN ORDER TO REMOVE
- The patient has undergone a urological procedure during this hospitalization
- The patient has had trauma to the urogenital area
- ICU patient with need for monitoring accurate I&O AND unable to use urinal or bedpan
- Physician documented urinary retention - please see Nurse Driven Urinary Retention Protocol
- Physician documented urinary obstruction
- Open stage III or greater wound in sacral or perineal area AND patient has urinary incontinence
- Strict prolonged immobilization due to multiple trauma or surgery

If the patient does not meet the daily criteria above for continued Foley placement, RNs can remove the Foley through the Nurse Driven Foley Catheter Protocol.

What is a Multi-Drug Resistant Organism (MDRO)?

Micro-organisms, predominantly bacteria, which are resistant to one or more classes of antimicrobial agents. These pathogens are frequently resistant to most available antimicrobial agents. The incidence of MDRO infections among hospital patients continues to rise, despite widespread efforts to control the spread. MDROs are even being seen among patients in the community.

Two main factors drive the spread of MDROs: Antibiotic misuse and overuse by clinicians & Cross-transmission between patients, usually in health care settings.

These highly resistant organisms include:

- VRE – Vancomycin-resistant enterococci
- CRE – Carbapenem-resistant enterobacterales
- ESBLs – Extended Spectrum Beta Lactamase (E. coli and Klebsiella)
- MRSA – Methicillin/Oxacillin-resistant S. aureus (also resistant to cephalosporins)
- Acinetobacter baumannii
- Stenotrophomonas maltophilia
- Burkholderia cepacia
- Strep pneumonia resistant to Penicillin
- Pseudomonas aeruginosa resistant to Imipenem
- Any gram negative bacilli resistant to 3rd generation cephalosporins, quinolones, and aminoglycosides
- Multi-drug resistant TB
- Clostridium difficile

Impact of Multi-Drug Resistant Organism on the Health Care System

Infections caused by Multi-Drug Resistant Organisms (MDROs) are associated with poor clinical outcomes, including an increased risk of death. They are also associated with significant increase in costs to hospitals, mostly attributable to an increase in length of stay for patients. Media, legislative, and regulatory pressures are compelling hospitals to more effectively control Multi-Drug Resistant Organisms (MDROs)

Strategies to Prevent the Spread of Multi-Drug Resistant Organisms (MDROs)

Hand hygiene is the single most important tool to prevent Multi-Drug Resistant Organism (MDRO) transmission in hospitals:

Use of alcohol-based hand disinfection products greatly improves compliance among clinicians.

Hand hygiene must be performed after removal of gloves. Gloves can have microscopic holes that may allow pathogens to reach the skin.

Even well-resourced organizations struggle to maintain hand hygiene compliance near 100%.

Isolation precautions (the use of gowns and gloves) is the standard of care for preventing transmission from patients with Multi-Drug Resistant Organism (MDRO)

Wearing gowns and gloves when there is direct physical contact with the patient and/or objects in their room prevents transmission of Multi-Drug Resistant Organisms (MDROs).

The use of gowns is much less expensive than the cost of treating an infection caused by an Multi-Drug Resistant Organism (MDRO).

Gowns must not be re-used. Re-use can lead to contamination of health care worker's clothes and an increased risk of transmission.

Compliance is an ongoing challenge.

Environmental hygiene is aimed to eliminate the reservoir of Multi-Drug Resistant Organisms (MDROs) on hospital surfaces and equipment

Multi-Drug Resistant Organisms (MDROs) can persist in the environment despite routine cleaning practices.

Some Multi-Drug Resistant Organisms (MDROs) can remain for weeks without proper surface disinfection.

Patients admitted to rooms previously occupied by a patient with an Multi-Drug Resistant Organism (MDRO) are at higher risk for developing infection with that Multi-Drug Resistant Organism (MDRO).

If you have any questions or concerns, contact an Infection Preventionist by calling extension (256) 265-8094.

INFLUENZA

As you know people get sick and die from the flu every year. It is estimated that, on average 5% to 20% of U.S. residents get the flu, and more than 200,000 people are hospitalized.

Flu is spread by droplets from coughing or sneezing of infected persons. It can also be spread by touching a surface contaminated with the virus, then touching the eyes, nose or mouth. Adults may spread influenza to others from the day before developing symptoms to approximately five days after onset. Influenza is diagnosed by clinical symptoms and or diagnostic testing.

Vaccination is the best way to reduce the chances of getting ill from influenza and from giving it to patients, co-workers and your family. The vaccine is safe, effective and free to physicians and allied health members through the Employee Occupational Health Department. Use of anti-viral medication can be helpful to treat and prevent influenza. Preventive measures such as respiratory etiquette, washing your hands and staying home when you are sick are important steps that help stop the spread of germs.

Health care workers must take the lead in influenza prevention. Many professional organizations are calling for 100% vaccination of health care workers, stating it is ethically justified, necessary and long overdue, including: The American Academy of Pediatrics, The Association of Professionals in Infection Control (APIC), The Infectious Disease Society of America, The American Hospital Association and The American Medical Association.

Please complete and return the card in the back of this education guide verifying whether you have or have not received your influenza vaccination this season. The Centers for Medicare & Medicaid Services (CMS) and the Centers for Disease Control and Prevention (CDC) require that hospitals report the rate of vaccination among the Medical Staff and all other health care team members. Your assistance with prompt reporting is greatly appreciated.

TRANSFUSION ASSOCIATED CIRCULATORY OVERLOAD (TACO)

As mandated by the AABB, the purpose of this education is to aid in prevention of Transfusion Associated Circulatory Overload (TACO).

TACO accounted for 22% of transfusion related fatalities reported to FDA from 2010-2014. Included are risk factors, diagnostic criteria and clinical features of TRALI versus TACO. Since many of the risk factors and/or symptoms are clinical in nature, the responsibility for the determination of whether the recipient is at risk for TACO falls predominantly on the ordering physician or the authorized health professional. Risk factors include:

- Increasing age
- Medical conditions: congestive heart failure; chronic renal failure
- Transfusion: infusion rate; total volume transfused; positive fluid balance prior to transfusion

Once the risk is known, the recommendation is to follow some or all of the following actions to aid in prevention or in recognition of TACO:

- Avoid unnecessary transfusion.
- Slow the rate of transfusions.
- Select the lowest volume components (inform blood bank a low volume product is requested).
- Avoid concurrent crystalloid infusion.
- Consider prophylactic diuretic therapy.
- More frequent monitoring of at risk patients during and immediately after transfusion.

Symptoms of TACO (usually within 6 hours of cessation of transfusion) include:

- Acute respiratory distress/dyspnea
- Positive fluid balance
- Left heart failure
- Elevated central venous pressure
- Radiographic evidence of (worsening) pulmonary edema
- Elevated BNP

If symptoms of TACO **develop**:

- Stop the transfusion, as indicated, and initiate transfusion work up
- Consider chest X-ray and BNP
- Consider diuretic therapy

TACO can be confused with TRALI (transfusion related acute lung injury) as both can have dyspnea, pulmonary infiltrates and rales. Differences include:

	TACO	TRALI
TEMPERATURE	Often unchanged	Often elevated
BLOOD PRESSURE	Hypertension	Hypotension
PULM ARTERY PRESSURE	>18mmHg	<18mmHg
DIURETIC RESPONSE	Significant	Minimal
PULMONARY FLUID	Transudate	Exudate

MATERNAL HEMORRHAGE – WHY THIS MATTERS IN THE ED

Maternal hemorrhage may present postpartum (up to 6 weeks) or after delivery at another facility. Patients may present for non-OB complaints (dizziness, syncope, abdominal pain). Delays in recognition and OB activation increase mortality.

Think Hemorrhage If:

- Recent pregnancy or delivery (vaginal or C-section)
- Heavy vaginal bleeding or clots
- Syncope, weakness, pallor
- Tachycardia or hypotension (late sign)
- Decreasing urine output

Normal vitals do NOT exclude significant blood loss

Ed Priorities – First 10 Minutes

1. Activate Maternal Hemorrhage Protocol
2. Call OB immediately
3. Place patient on continuous monitoring
4. 2 large-bore IVs
5. Begin QBL (weigh pads, measure suction)
6. Type & cross, notify blood bank early

Initial Management

- Uterine massage if uterus palpable/boggy
- Oxytocin IV/IM (first-line if available)
- TXA early if ongoing bleeding (within 3 hrs of birth)
- Treat reversible causes while awaiting OB

Remember The “4 Ts”

- Tone – Uterine atony (most common)
- Trauma – Lacerations, rupture
- Tissue – Retained products
- Thrombin – Coagulopathy

Massive Transfusion – Think Early

Activate MTP for:

- Ongoing bleeding + instability
- Rapid drop in BP or mental status
- Poor response to fluids

Use balanced blood products

Warm patient and blood

ED Key Takeaways

- Postpartum = high-risk window
- Treat bleeding before hypotension
- Early OB involvement saves lives
- Do not wait for labs to transfuse if unstable

Reference: Obstetric Management of Postpartum Hemorrhage

Patient education is available in the postpartum discharge book and inside 1Chart.

MATERNAL HEMORRHAGE — WHY THIS MATTERS ON L&D

- Hemorrhage is predictable, preventable, and treatable
- Most maternal deaths involve delayed escalation
- Standardized response improves outcomes

Definition of Postpartum Hemorrhage (PPH):

- $\geq 1,000$ mL blood loss OR
- Any blood loss with signs of hypovolemia

Use Quantified Blood Loss (QBL) for all births

Immediate L&D Response

Recognize early – act fast

1. Activate Hemorrhage Protocol
2. Call for help (OB, anesthesia, charge RN)
3. Continuous vitals q5 min
4. Uterine massage
5. 2 large-bore IVs
6. Draw labs (CBC, fibrinogen, coags)

Management By Cause – “4 Ts”

TONE (Uterine Atony – Most Common)

- Oxytocin IV/IM
- Methylergonovine (avoid HTN)
- Carboprost (avoid asthma)
- Misoprostol
- TXA early if bleeding persists

Trauma

- Inspect cervix/vagina/perineum
- Repair lacerations promptly

Tissue

- Manual placental removal
- Ultrasound if retained products suspected

Thrombin

- Correct coagulopathy
- Replace fibrinogen early

Escalation

- Bimanual compression
- Intrauterine balloon tamponade
- Activate Massive Transfusion Protocol
- Surgical intervention as indicated

Post-Hemorrhage Care

- Continue close monitoring
- Repeat labs
- Debrief team
- Document QBL & response
- Patient education before discharge

L&D Key Takeaways

- QBL > estimation
- Uterine atony is most common—treat first
- Escalate early
- Teamwork and protocols save lives

Reference: Obstetric Management of Postpartum Hemorrhage

Patient education is available in the postpartum discharge book and inside 1Chart.

MATERNAL SEVERE HYPERTENSION & PRE-ECLAMPSIA — WHY THIS MATTERS IN THE ED

- Preeclampsia is often missed in the ED, especially postpartum or when patients present for non-obstetric complaints.
- One severe BP is a true emergency—delays increase stroke and maternal death risk.

Think Preeclampsia If:

- Pregnant ≥ 20 weeks OR ≤ 6 weeks postpartum
- Chief complaints:
 - Headache, vision changes
 - RUQ/epigastric pain
 - SOB, chest pain
 - Seizure, AMS
 - “Just doesn’t feel right”

Severe Hypertension = Emergency

- SBP ≥ 160 OR DBP ≥ 110
- Do NOT wait for repeat BP or labs to treat

Immediate Ed Actions (First 30–60 Min)

1. Repeat BP manually (confirm severity)
2. Start antihypertensive treatment
3. Initiate Magnesium Sulfate if preeclampsia suspected
4. Consult OB immediately
5. Continuous BP & pulse ox monitoring

First-Line BP Treatment

Use what is available fastest:

- IV Labetalol
20 mg $\rightarrow$ 40 mg $\rightarrow$ 80 mg IV q10 min (max 220 mg)
- IV Hydralazine
5–10 mg IV q20 min
- PO Nifedipine IR
10–20 mg PO, repeat in 20 min

Goal: SBP < 160 , DBP < 110

Seizure Prophylaxis

Magnesium Sulfate

- 4–6 g IV load $\rightarrow$ 1–2 g/hr infusion
- Give even before OB arrival if concern is high

Labs (DO NOT DELAY TREATMENT)

- CBC, CMP, AST/ALT
- Creatinine
- Urine protein/creatinine (if time allows)

Key ED Reminders

- Postpartum patients are high risk
- Preeclampsia may present without proteinuria
- Headache + hypertension = assume severe disease

MATERNAL SEVERE HYPERTENSION & PRE-ECLAMPSIA — WHY THIS MATTERS ON L&D

- Most preventable maternal strokes occur due to delayed treatment of severe BP
- Standardized response saves lives

Diagnosis (ACOG)

Severe Hypertension

- SBP ≥ 160 OR DBP ≥ 110 (single reading sufficient)

Preeclampsia with Severe Features

- Severe BP OR
- Platelets $< 100K$
- Creatinine > 1.1 or doubling
- AST/ALT $\geq 2\times$ normal
- Pulmonary edema
- Persistent headache or visual symptoms

Immediate L&D Response

Treat within 30–60 minutes

- Activate Severe HTN protocol
- Place patient on continuous monitoring
- Start antihypertensive therapy
- Initiate Magnesium Sulfate
- Draw labs (do not delay meds)

First-Line Antihypertensives

- IV Labetalol
- IV Hydralazine
- PO Nifedipine IR

Maintain BP $< 160/110$

Magnesium Sulfate

- 4–6 g IV loading dose
- 1–2 g/hr maintenance
- Continue intrapartum & 24 hrs postpartum

Monitor:

- Reflexes
- Respiratory rate
- Urine output

Delivery Considerations

- Severe preeclampsia = maternal stabilization first
- Route and timing of delivery based on:
 - Gestational age
 - Maternal/fetal status
- BP control is NOT a reason to delay delivery

Postpartum Management

High risk period for stroke & eclampsia

- BP monitoring for ≥ 72 hrs
- Treat persistent BP $\geq 150/100$
- Discharge education mandatory
- BP check within 7–10 days

Key L&D Takeaways

- One severe BP = treat now
- Magnesium prevents seizures
- Postpartum vigilance saves lives
- Standardization reduces mortality

Reference: Magnesium Sulfate for Pregnancy Induced Hypertension Policy

Patient education is available in the postpartum discharge book and inside 1Chart.

CLICK HERE

to submit notice of receiving
and reviewing current provider
educational requirements.